

The coppe document class

Version 3.8

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Abstract

In this work, it is described the `coppe` document class as well as other files distributed by the COPPETEX project. This class is suitable for writing academic dissertations, thesis and qualifying exams according to the formatting rules of the Alberto Luiz Coimbra Institute for Graduate Studies and Research in Engineering. The minimalist set of macro commands allows its users to concentrate most of their efforts on text composition rather than on the document layout.

1 Introduction

Writing documents in L^AT_EX may be a laborious task when the authors have to prepare their manuscripts rigorously respecting formatting rules imposed by publishers. Regardless of difficulty, a lot of thesis presented to the Coordination of Graduate Studies and Research in Engineering of the Federal University of Rio de Janeiro (COPPE/UFRJ) is typesetted in L^AT_EX. This demand motivated the creation of the COPPETEX project, which tries to facilitate and encourage the use of L^AT_EX within the COPPE/UFRJ scope.

The `coppe` document class is the main product of COPPETEX. It was designed to be clear and succinct. It enables the creation of dissertations, qualifying exams and thesis in a simple and automatic way. The main goal of the `coppe` class is to maintain authors strictly focused on text composition without worrying about margins sizes, line spacing, paper size, vertical and horizontal alignment, etc. The COPPETEX project comprehends also BibTEX and MakeIndex style files for creating lists of references, symbols and abbreviations. Although there aren't official guidelines to write qualifying exams, we provide this option just for convenience, as this exam is a requisite to obtain the DSc degree and, for some of the programs, the MSc degree.

In which follows, it is described the user interface of the `coppe` class. Some details about using the style files cited above are also given. We use the term *thesis* to generally refer to dissertation, qualifying exam, and thesis itself.

2 License

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copying, distributing or modifying the source code, among other acts covered by this license.

To see the full text of the GNU GPL license, go to the `COPYING` file attached to this package.

3 Support

We maintain a mailing list where users can send questions, comments, and bugs to. More details can be found [here](#).

However, as the project maintenance was transferred to a new repository, bug reports, as well as new feature requests, should be directed to <https://github.com/COPPE-UFRJ/CoppeTeX>.

4 User interface

`\frontmatter` A thesis to be approved by the Academic Registry at COPPE/UFRJ must contain three-parts: *front*, *main* and *back* matters (COPPE/UFRJ Academic Registry guidelines). Each one of these parts is started by calling its corresponding macro `\frontmatter`, `\mainmatter` or `\backmatter`. The front matter of a thesis consists of front cover and face, cataloging page, dedication, acknowledgments, abstracts, table of contents, and lists of tables, algorithms, symbols and abbreviations. The main matter is just composed by chapters, while the back matter usually consists of bibliographic references, appendices and index.

You must invoke the `\frontmatter` macro immediately after the `\maketitle` one. The `\mainmatter` command comes right before the first chapter, and `\backmatter` must be typed before the list of references.

Cover (capa)

The ABNT cover suggested in the manual's Annex A. The `\maketitle` command builds it automatically, ahead of the title page: the university, the full COPPE name, and the graduate program (taken from `\department`), followed by the author, the title, and the city and year, all centred and in uppercase. The UFRJ and COPPE logos head the page. The capa is not counted.

Front face

The front face is unnumbered. There, it is not allowed to use hyphenation (COPPE/UFRJ Academic Registry guidelines). It is constructed by calling `\maketitle`. Next, it is described the commands used to enter the information required to create it.

`\author` The `\author` command was redefined. Here, it takes two arguments: the author's first names and surname, e.g., `\author{First Names}{Surname}`. The words should be typed with only first letters in uppercase.

`\title` The macros `\title` and `\foreigntitle` are used to enter the titles of your monograph in Brazilian Portuguese and English respectively (the two built-in languages of CoppeTeX 4.x). For documents whose main language is neither

brazilian nor english – e.g. spanish, french, italian, or a user-supplied language –, also call `\titlein{<lang>}{<text>}` to register the title in that language; the cover and folha de rosto print whichever title matches the main language declared in the class option. See the section “Multilingual support” below for the complete picture. The `babel` package is loaded automatically by `coppe.cls`, so you do not need to load it yourself.

`\advisor` Every COPPE student is coordinated by at least one advisor. M.Sc. and D.Sc. students can have at most 2 and 3 advisors, respectively. Their names must be provided by issuing the command `\advisor` as below:

```
\advisor{Title}{Advisor's Name}{Surname}{Degree}
\advisor{Title}{Second Advisor's Name}{Surname}{Degree}
\advisor{Title}{Third Advisor's Name}{Surname}{Degree}
```

The advisors are not necessarily members of the thesis examination board. Thus, it is required to enter the names of all examiners using the `\examiner` macro. The examiners' names are entered differently:

```
\examiner{Title}{First Examiner's Name Surname}{Degree}
\examiner{Title}{Second Examiner's Name Surname}{Degree}
...
\examiner{Title}{N-th Examiner's Name Surname}{Degree}
```

Remember that all names must be given before calling `\maketitle`.

`\department` The Alberto Luiz Coimbra institute is divided into 12 academic units: Biomedical Engineering (PEB), Civil Engineering (PEC), Electrical Engineering (PEE), Mechanical Engineering (PEM), Metallurgical and Materials Science Engineering (PEMM), Nuclear Engineering (PEN), Ocean Engineering (PENO), Energy Planning (PPE), Production Engineering (PEP), Chemical Engineering (PEQ), Systems Engineering and Computer Science (PESC), and Transportation Engineering (PET). You must specify your department using one of the above abbreviations, e.g., `\department{PEC}`.

`\date` This macro is used to set the month and year of defense. This information is required to create the front face, cataloging details page and abstracts. For example, October 2007 should be entered as `\date{10}{2007}`.

`\keyword` The keywords should describe the concentration areas of your work. You must provide them as follows:

```
\keyword{First Keyword}
\keyword{Second Keyword}
...
\keyword{N-th Keyword}
```

Usually, six words are enough.

Cataloging details

This page contains cataloging information useful for librarians. Fortunately, it is automatically generated from the data you entered at the time you call `\maketitle`. It is not needed in qualifying exams, though.

Dedication (optional)

`\dedication` This macro was added for convenience. The input text is placed at the right bottom of a blank page. It is emphasized and in normal size.

Abstracts

`abstract (env.)` As stated by the COPPE/UFRJ Academic Registry, abstracts must be in one page each, with at most 250 words. We recommended that they should be only one paragraph long. They must be defined inside the environments `abstract` and `foreignabstract`.

Lists of symbols and abbreviations (optional)

`\abbrev` The lists of symbols and abbreviations are optional, although highly recommended.
`\syml` It is a good practice to define a symbol/abbreviation in its first occurrence in the text. To define a symbol use `\syml[alphabetic symbol]{Symbol}{Symbol Definition}`, and for abbreviations `\abbrev[alphabetic symbol]{Abbreviation}{Abbreviation Definition}`. These commands are called *dummy*, since they don't output anything at the place they are executed, just an entry in the correspondent list.
`\makeloabbreviations` These lists are lexicographically sorted by using the MakeIndex program, which
`\makelosymbols` is part of any L^AT_EX implementation. For `\syml`, if the optional parameter is
`\printloabbreviations` provided, it will used as sort key. This was later, in 2024, implemented also for
`\printlosymbols` `\abbrev`, otherwise `Symbol`, or `Abbreviation` will be used as sort key, what can result in an undesirable order if it contains L^AT_EX commands, mathematical symbols, or mix of uppercase and lowercase. MakeIndex needs two commands to create a final sorted list: one which generates a list of entries and the other that indicates the position where the list will be printed out. To generate the lists of symbols and abbreviations, the `coppe` class provides the commands `\makeloabbreviations` and `\makelosymbols`, respectively. They must be called in the document preamble. The commands `\printlosymbols` and `\printloabbreviations` have to be invoked at the point where you want these lists appear, e.g., following the list of tables as showed in the example. Once you call `latex`, it will be created two files with extensions `abx` and `syx`, which contain MakeIndex input data. They must be processed with `makeindex` in order to get the lists correctly produced, redirecting the output to files with extension `lab` and `los` respectively:

```
makeindex -s coppe.ist -o example.lab example.abx
makeindex -s coppe.ist -o example.los example.syx
```

Note the `-s` option for specifying the style `coppe.ist`. Now, rerun `latex` twice to get the references solved and you are done.

References

COPPE_T_E_X produces citations and the reference list with `biblatex` and the `biber` backend, through its own bibliography style implementing the post-2020 ABNT NBR 6023 rules. The class loads this style automatically: there is no `\bibliographystyle` to set and no Bib_T_E_X `.bst` file to choose (the former `coppe-plain` and `coppe-unsrt` Bib_T_E_X styles have been retired).

By default, citations follow the author–date style, e.g., “Chartier (2002)”. Passing the `numbers` class option switches to the numeric style, in which citations are bracketed numbers and the reference list is numbered in order of citation (unsorted).

Declare your bibliography database in the preamble with `\addbibresource` and print the list where it should appear with `\printbibliography`:

```
\addbibresource{example.bib}    % in the preamble
...
\printbibliography              % where the list goes
```

Entry types and fields may be written in English (`@book`, `title`, `author`) or with their unaccented Portuguese synonyms (`@livro`, `titulo`, `autor`).

Run in sequence `LATEX`, `biber`, and twice again `LATEX` to resolve the citations and references. The style is `natbib` compatible (via the `biblatex natbib=true` option), so you may freely issue `\citet` and `\citep`, as well as the other `natbib` commands.

Reference entry types

Every ABNT document category maps to a `biblatex` entry type, each with an unaccented Portuguese synonym (a worked example of every type is in the sample manuscript’s reference list):

Academic `@book` (`@livro`); `@article` (`@artigo`); `@artigodejornal` (newspaper); `@incollection` (`@partedelivro`); `@inbook` (`@emlivro`); `@inreference` (`@verbete`); `@proceedings` (`@anais`); `@inproceedings` (`@trabalhodeevento`); `@thesis` (`@tese/@dissertacao`, with the presets `@dissertacaomestrado`, `@tesedoutorado`, `@teseph`, `@tcc`); `@report` (`@relatorio`); `@periodical` (`@periodico`); `@manual`; `@misc` (`@diverso`); `@online`.

Special (NBR 6023 §4.2.5–4.2.14) `@patent` (`@patente`); `@legislation` (`@legislacao/@atonormat`); `@jurisdiction` (`@jurisprudencia`, with a `relator` field); `@legal` (`@documentojuridico`); `@standard` (`@norma`); `@music` (`@partitura`); `@map` (`@mapa`); `@image` (`@iconografico`); `@artwork` (`@obradearte`); `@video` (`@audiovisual`); `@movie` (`@filme`); `@letter` (`@correspondencia`); `@dataset` (`@conjuntodedados`).

Games and TV `@game` (`@jogo`); `@boardgame` (`@jogodetabuleiro`); `@videogame` (`@jogoeletronico`); `@tvshow` (`@programadetv`).

When an entry has no `author` or `editor`, it is entered by title, with the first significant word (and a leading article) in capitals.

Appendix and Annex (Optional)

`\appendix` Appendices and annexes are optional chapters that are part of the back matter.

`\annex` The `\appendix` command is a standard `LATEX` command used before all appendices. `COPPELTEX` introduces the `\annex` command, which should be used only in the back matter (i.e., after the `\backmatter` command) and only after all existing `\appendix` chapters. This restriction is due to implementation constraints.

Therefore, the order for backmatter is:

```
\backmatter
\printbibliography
\appendix
\chapter{An Appendix}
\chapter{Another Appendix}
\annex
\chapter{First Annex}
\chapter{Second Annex}
```

Annex was introduced in v3.5

Final colophon page (`\coppetexfinalpage`)

`\coppetexfinalpage` Optional closing page modelled on the printer’s *colophon*: it records, in Portuguese, the tooling and edition used to typeset the thesis. Call it once, after every other piece of content, immediately before `\end{document}`. The macro

- forces a verso (even) page, so in a two-sided print the colophon sits opposite the rear cover;
- types the body at the foot of the page (`\vfill`);
- renders the text in Portuguese unconditionally (Brazilian `babel`), even when the thesis main language is English, Spanish, French or Italian — the colophon is institutional metadata, not body content.

Three slot macros are user-overridable. Redefine them with `\renewcommand` before `\coppetexfinalpage` if the document deviates from the defaults:

`\coppefinalengine` the T_EX engine identifier. Default: auto-detected via `iftex` (`\hologo{pdfLaTeX}`, `\hologo{LuaLaTeX}` or `\hologo{XeLaTeX}`).

`\coppefinalfont` the text body font. Default: *Computer Modern (padrão L^AT_EX)*. Redefine if you loaded `lmodern`, `mathptmx`, an OpenType math font, etc.

`\coppefinalmanual` the reference manual title (typeset in italics). Default: the current COPPE/UFRJ norm. Redefine if your work follows an earlier edition.

The macro also prints the COPPE_TE_X class version (captured into `\coppe@version` at class-load time, so it survives any later package that may redefine the global `\fileversion`) and the GitHub repository URL, <https://github.com/COPPE-UFRJ/CoppeTeX>.

5 Class options

There are some options users can specify in order to customize the appearance of the output produced by the `coppe` class. These options can be passed to `coppe` as follows: `\documentclass[option1, option2]{coppe}`. In which follows, we give a brief description of all supported options.

dsc, msc, dsccexam, msccexam The `coppe` class is able to produce thesis, dissertations, and qualifying exams, which are enabled by the `dsc`, `msc`, `msccexam`, and `dsccexam` options, respectively.

doublespacing The default line spacing is one-and-a-half. For enabling double spacing between lines, use the `doublespacing` option.

numbers The default citation style is the author–date scheme. For numbered citations, specify the `numbers` option to the `coppe` class; the reference list is then numbered in order of citation (unsorted). The matching `biblatex` style is selected automatically in either case.

brazilian, english, spanish, french, italian Pick the document’s *main* language. The class loads Babel with that language as the main one (the last in the Babel option list); inside the document `\selectlanguage{...}` and `\begin{otherlanguage}{...}` keep working as usual. The default, when no language option is given, is **brazilian**.

- **brazilian** and **english** are *built-in*: their strings ship inside `coppe.cls` and need no extra file.
- **spanish, french, italian** are shipped as *language packs*: two files per language (`coppe-lang-<lang>.def` and `<lang>-coppe.lbx`) that must sit next to `coppe.cls` (or in your project directory). They are auto-loaded by the class. Picking one of them as the main language without the matching pack files installed raises a clear class error.
- Any other Babel-known language can be plugged in by writing the same two files and either passing the language name as a class option or calling `\usecoppelanguage{<lang>}` in the preamble.

See the section “Multilingual support” below for the three-slot architecture, the `\titlein` / `\brazilianabstract` API, and how to create a new language pack.

5.1 Changing document identification

`\freeconfig` The user could *optionally* use the command `freeconfig` to modify the parameters that print the document identification. The command `freeconfig` needs all those parameters, which are degree initials, degree name, title, foreign title, local doctype, and foreign doctype as in the following example:

```
\freeconfig{Dr.}{Philosophiae Doctor}{PhD}{Doutor}{Dissertation}{Tese}
```

5.2 Multilingual support

CoppeTeX 4.x reserves three fixed language slots:

main the language of the body, captions, TOC labels and the `abstract` environment. Picked by the class option (**brazilian** by default; one of **english, spanish, french, italian** also accepted out of the box; any other Babel language possible with a user-supplied pack).

foreign the language of the `foreignabstract` environment. `english` by default for every main language; switches automatically to `brazilian` when `main = english`.

third optional. Loaded on demand with `\usecoppelanguage{<lang>}` in the preamble, useful when the body quotes or cites in a third language (e.g. a Spanish-main thesis citing a French source). Does not have a class option of its own.

`\titlein`
`brazilianabstract (env.)`
`\usecoppelanguage`

Two pieces of user-facing API come with the multilingual model:

- `\titlein{<lang>}{<text>}` records the title in any language. `\title{...}` still writes the Brazilian-Portuguese title and `\foreigntitle{...}` still writes the English one (back-compat); for a main language outside the `brazilian/english` pair, also call `\titlein{<lang>}{...}`. The cover and folha de rosto print whichever title matches `\coppe@mainlang`.
- The `brazilianabstract` environment is the optional third abstract used when the main language is neither `brazilian` nor `english` – typically a Spanish/French/Italian thesis that also wants a Portuguese resumo for UFRJ examiners. It wraps its body in `\begin{otherlanguage}{brazilian}` and uses the Portuguese template strings. A document with `main = brazilian` can simply keep using `abstract`; `brazilianabstract` is redundant then.
- `\usecoppelanguage{<lang>}` loads an extra language pack in the preamble without making it the main or foreign language.

Institutional names stay Portuguese

The cover, folha de rosto and ficha catalográfica are a Brazilian institutional template (NBR 14724 + COPPE manual). The strings `universityname`, `cityname`, `statename`, `countryname`, the departments declared with `\department` and the surrounding “apresentada/submetida ao...” wording are deliberately kept in Portuguese for every main language. Only the title (printed via `\coppe@selecttitle`) follows the main language.

Writing a new language pack

A language pack is just two files dropped next to `coppe.cls` (or anywhere on the `TEXINPUTS` path):

`coppe-lang-<lang>.def` class-level strings. Populate the dispatcher table with `\copperdefstring{<lang>}{<key>}{<value>}` for every key already populated for `brazilian/english` in `coppe.cls` (appendix, annex, frame, listframe, source, program, listprogram, references, in:, art, platform, directedby, producedby, listabbreviation, listsymbol, glossary, advisor, advisors, dept, approvedmale, approvedfemale, universityname, cityname, statename, countryname, monthname1...monthname12, algorithm2eopt, babelname). Finish with

```
\AtBeginDocument{\DeclareLanguageMapping{<lang>}{<lang>-coppe}}
```

so the `biblatex` mapping is registered after the bibliography engine is loaded by `coppe.cls` (which happens *after* the language pack is read).

`<lang>-coppe.lbx` `biblatex` localization strings. `\InheritBibliographyExtras{<lang>}` then `\DeclareBibliographyStrings` for the keys `availablefrom`, `urlseen`, `in`, `editor`, `editors`, `depositor`, `coppescale`, `mscdiss`, `dscthis`, `phdthesis`, `tcc`. The shipped `spanish-coppe.lbx`, `french-coppe.lbx` and `italian-coppe.lbx` are the reference templates to copy from.

Tip: the `algorithm2eopt` key holds whatever `algorithm2e` calls the language. Mind the spelling: `algorithm2e` ships `portuguese`, `english`, `spanish`, `french` and `italiano` – not `italian`.

Example: a Spanish-main thesis

```
\documentclass[spanish,dsc]{coppe}
\addbibresource{example.bib}
\begin{document}
  \title{T\`i tulo em portugu\~es}      % Brazilian abstract slot
  \foreigntitle{English title}          % English abstract slot
  \titlein{spanish}{T\`i tulo en espa\~nol} % cover/folha
  \author{Juan}{P\`erez}
  \advisor{Prof.}{Mar\`a}{Garc\`a}{D.Sc.}
  \department{PESC}\date{01}{2026}
  \keyword{multilenguaje}
  \maketitle
  \frontmatter
  \begin{abstract}      Resumen principal en espa\~nol. \end{abstract}
  \begin{foreignabstract} English foreign abstract. \end{foreignabstract}
  \begin{brazilianabstract} Resumo em portugu\~es para a banca. \end{brazilianabstract}
  \tableofcontents
  \mainmatter
  \chapter{Introducci\`on} Cuerpo en espa\~nol\dots
\end{document}
```

6 Files in your project folder

In a normal install where CoppeTeX is part of your `TEXINPUTS` path (e.g. inside the local `texmf` tree of MiKTeX or TeX Live), you do not need to think about this list at all — the files travel with the class. This section is for two situations where you do:

- shipping the class together with your manuscript (a self-contained submission package); or
- installing CoppeTeX by hand, dropping the files next to `coppe.cls` in the project directory.

Always required

These files must be findable at every compilation, regardless of the main language or any class option:

`coppe.cls` the document class itself.

`coppe.dbx` custom `biblatex` datamodel (declares the ABNT-specific entry types `@game`, `@videogame`, `@boardgame`, `@tvshow` and `@map`, plus a few custom fields).

`coppe.bbx` and `coppe.cbx` `biblatex` bibliography and citation styles for the author–date scheme (the default).

`brazilian-coppe.lbx` and `english-coppe.lbx` `biblatex` localisation tables for Portuguese and English. Both are required *even when neither language is the main one*, because `coppe.bbx` registers

```
\DeclareLanguageMapping{brazilian}{brazilian-coppe}
\DeclareLanguageMapping{english}{english-coppe}
```

unconditionally, so `biblatex` tries to read both LBX files at every compilation.

Conditional on class options

`coppe-numeric.bbx` and `coppe-numeric.cbx` only when the `numbers` class option is given (`\documentclass[numbers]{coppe}`). Without it, `biblatex` never tries to read them.

`coppe.ist` only when the glossary, list-of-symbols, or list-of-abbreviations features are used. It is the `makeindex` style file invoked by the `makeindex -s coppe.ist ...` step described in subsection *Lists of symbols and abbreviations* above.

Conditional on the main language

The main language is chosen by the matching class option (`brazilian`, `english`, `spanish`, `french`, `italian`). The two built-in languages ship their strings inside `coppe.cls` and so need no extra file. The three language packs add two files each — one class-level (`coppe-lang-<lang>.def`) and one `biblatex`-level (`<lang>-coppe.lbx`):

Main language	Extra files needed
<code>brazilian</code> (default)	— (built into <code>coppe.cls</code>)
<code>english</code>	— (built into <code>coppe.cls</code>)
<code>spanish</code>	<code>coppe-lang-spanish.def</code> , <code>spanish-coppe.lbx</code>
<code>french</code>	<code>coppe-lang-french.def</code> , <code>french-coppe.lbx</code>
<code>italian</code>	<code>coppe-lang-italian.def</code> , <code>italian-coppe.lbx</code>

Loading a third language at preamble time via `\usecoppelanguage{<lang>}` brings in the same pair of files for that language. A user-supplied language pack follows the same scheme: drop `coppe-lang-<lang>.def` and `<lang>-coppe.lbx` next to `coppe.cls`.

Cover assets

The institutional title page produced by `\maketitle` needs two PDF logo files at compile time:

`ufrj-logo.pdf` and `coppe-logo.pdf` the UFRJ shield and the COPPE/UFRJ wordmark. Both ship with the CoppeTeX distribution; they must be findable from the project directory (or on `TEXINPUTS`).

7 Required L^AT_EX packages

The class is built on the standard book class (`12pt`, `a4paper`, `twoside`) and brings in the stock CTAN packages listed below with `\RequirePackage`. On a full T_EX Live or MiK_TE_X install they are all present already, so this list matters only for a minimal install — or when you simply want to know which commands are already in scope. You do *not* need to `\usepackage` any of these yourself; every command and environment they provide is available in your manuscript. Options in brackets are the ones the class passes.

Engine and encoding

`iftex` detects the running engine, so the class can branch its setup between pdfT_EX and the Unicode engines (LuaT_EX/XeT_EX).

`inputenc` (`utf8`) UTF-8 input; loaded *only under pdfT_EX* (the Unicode engines read UTF-8 natively).

`fontenc` (`T1`) T1 font encoding, for correctly composed accented glyphs and hyphenation.

Mathematics and symbols

`amsmath` the AMS mathematics environments.

`amssymb` the AMS symbol fonts; loaded *only under pdfT_EX*. Under a Unicode engine load `unicode-math` (or `amssymb`) yourself — the class warns at end of document if neither is present.

Page layout and spacing

`geometry` A4 two-side margins, binding offset and header height.

`fancyhdr` running headers and footers.

`titlesec` chapter and section title formatting.

`setspace` one-and-a-half line spacing (double with the `doublespacing` option).

Lists, tables and rules

`enumitem` customisable list spacing and labels.

`tabularx` tables with automatic-width columns.

`booktabs` professional horizontal rules (`\toprule`, `\midrule`, `\bottomrule`).

`longtable` tables that break across pages.

Floats, graphics and captions

`graphicx` image inclusion.

`float` float placement control (e.g. the `[H]` specifier).

`caption` caption typography (the ABNT caption-on-top layout).

`subcaption` subfigures and subtables.

Code, verbatim and algorithms

`listings` source-code listings.

`fancyvrb` extended verbatim.

`algorithm2e` (`linesnumbered`, `lined`, `algochapter`, `ruled`) typeset algorithms;
the natural-language option tracks the main language.

Bibliography, quotations and language

`babel` multilingual typesetting; the language options are computed from the main
and foreign language slots.

`csquotes` context-sensitive quotation marks (the recommended companion of
`biblatex`).

`biblatex` (`backend=biber`) the bibliography engine and the ABNT `coppe` styles
(`coppe` or, with the `numbers` option, `coppe-numeric`). The backend is
`biber`, not `bibTeX`.

Hyperlinks

`hyperref` clickable cross-references, PDF bookmarks and document metadata.

Low-level utilities

`etoolbox` e-TeX programming toolkit.

`ltxcmds` low-level command helpers (`\ltx@ifpackageloaded`, ...).

`ifthen` the `\ifthenelse` conditional.

`hyphenat` hyphenation control (`\nohyphens`, used on the cover).

`lastpage` `\pageref{LastPage}` for the cataloging page count.

hologo T_EX-family logos (`\hologo{LaTeX}, \dots`).

xcolor colour support (used by `listings` and `hyperref`).

8 Quick, useful tips

Pictures. The default picture format of L^AT_EX is the Encapsulated PostScript (EPS). If you use pdfL^AT_EX, the default format becomes the PDF, but you can equally load PNG files. For such, you must enter the name of your image file without extension, e.g., `\includegraphics{filename}`, and pdf_latex will firstly look for a file called `filename.pdf` and after for file `filename.png`. For producing high quality pictures with embedded fonts we recommend the Ipe drawing software available [here](#).

Fonts. The default font in L^AT_EX is the Computer Modern. If you would like to try its enhanced version, consider using the `lmodern` package. To use Times, it is recommended to load the package `mathptmx`, rather than the deprecated `times`. There is also an enhanced Times version available with the `tgtermes` package. You can still use the Arial font face with the package `uarial`.

Hyperref. When working with PDF's, there is the possibility to add extra information to the file as the author's name, document title, subject, keywords, etc. This is easily done with the `hyperref` package, which is loaded by the `coppe` class itself (since v4.0): the user must *not* load it again in the manuscript preamble. Author, title, subject and keyword metadata are filled in automatically by `\maketitle`.

Math (amssymb vs. unicode-math). The class always loads `amsmath`. The companion AMS symbol package `amssymb`, however, is loaded *only* under pdfL^AT_EX: it is a legacy 8-bit package built around the `msam/msbm` Type 1 fonts and clashes with `unicode-math`, the modern OpenType-math package used on Unicode engines (LuaL^AT_EX, XeL^AT_EX). On those engines the class skips `amssymb` and schedules an *end-of-document* check: if by then the author has loaded neither `unicode-math` nor `amssymb` (which they could have loaded by hand, though it is discouraged), a `ClassWarning` reminds them to add

```
\usepackage{unicode-math}
\setmathfont{Latin Modern Math} % or STIX Two Math, ...
```

to the manuscript preamble. The class deliberately does *not* auto-load `unicode-math`, because picking an OpenType math font is a typographic choice that belongs to the author. **Never load `amssymb` together with `unicode-math`:** it would re-establish the 8-bit AMS fonts on top of the OpenType math font and split symbols across two incompatible encodings.

Printing. To get your work correctly printed, you must ensure that any page scaling option (e.g., fit or shrink to printable area) isn't enabled. This kind of option often comes in print dialogs of document visualization softwares.

longquote Quotation To quote text larger than three lines, according to ABNT, you must increase the left margin to 4 cm, do not use quotation marks, and use a smaller font. The `coppe` class provides the `longquote` environment to easily make these adjustments.

9 A simple example

```

1 \example
2 \documentclass[dsc]{coppe}
3
4 \usepackage{rotating}          % rodando coisas, como tabelas
5 \usepackage[most]{tcolorbox} % caixas de texto
6 \usepackage{tikz}             % gráficos vetoriais (o tcolorbox já o traz)
7 % Sob LuaLaTeX/XeLaTeX, descomente as duas linhas a seguir para obter os
8 % símbolos AMS (\mathbb, \hbar, \varnothing, ...) via unicode-math. Sob
9 % pdfLaTeX, a classe já carrega amssymb automaticamente; deixe comentado.
10 \usepackage{unicode-math}
11 %\setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
12 \DeclareMathOperator{\RMSE}{RMSE} % operadores próprios (cap. de matemática)
13 \DeclareMathOperator{\diag}{diag}
14 % A classe coppe (v4.0) já carrega: amsmath, booktabs, longtable, subcaption,
15 % hyperref, biblatex, listings, fancyvrb, algorithm2e -- não recarregue esses
16 % pacotes aqui (em particular, hyperref deve ficar a cargo da classe para que
17 % a ordem com setspace e biblatex permaneça correta).
18
19 \addbibresource{example.bib}% base de referências (biblatex/biber)
20
21 \makelosymbols
22 \makeloabbreviations
23
24 \begin{document}
25   \title{Título da Tese}
26   \foreigntitle{Thesis Title}
27   \author{Nome do Autor}{Sobrenome}
28   \advisor{Prof.}{Nome do Primeiro Orientador}{Sobrenome}{D.Sc.}
29   \advisor{Prof.}{Nome do Segundo Orientador}{Sobrenome}{Ph.D.}
30   % A classe aceita de um a tres \advisor; se houver apenas um, o rotulo na
31   % capa e na folha de rosto sai como "Orientador" (singular), com dois ou
32   % mais sai como "Orientadores" (plural). Para o caso 1/3 orientadores,
33   % veja tests/test_brazilian_one_advisor.tex e tests/test_brazilian_three_advisors.tex.
34
35   \examiner{Prof.}{Nome do Primeiro Examinador Sobrenome}{D.Sc.}
36   \examiner{Prof.}{Nome do Segundo Examinador Sobrenome}{Ph.D.}
37   \examiner{Prof.}{Nome do Terceiro Examinador Sobrenome}{D.Sc.}
38   \examiner{Prof.}{Nome do Quarto Examinador Sobrenome}{Ph.D.}
39   \examiner{Prof.}{Nome do Quinto Examinador Sobrenome}{Ph.D.}
40   \department{PESC}
41   \date{01}{2024}
42
43   \keyword{Primeira palavra-chave}
44   \keyword{Segunda palavra-chave}
45   \keyword{Terceira palavra-chave}
46

```

```

47 \maketitle
48
49 \frontmatter
50 \dedication{A alguém cujo valor é digno desta dedicatória.}
51
52 \chapter*{Agradecimentos}
53
54 Gostaria de agradecer a todos os que contribu\iram para este trabalho:
55 orientadores, colegas, familiares e \as ag\ências de fomento. Lorem ipsum
56 dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt
57 ut labore et dolore magna aliqua.
58
59 \begin{abstract}
60
61 Apresenta-se, nesta tese, um exemplo de resumo. Lorem ipsum dolor sit amet,
62 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
63 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
64 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
65 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
66 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
67 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
68 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
69 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
70 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
71 aspernatur aut odit aut fugit.
72
73 \end{abstract}
74
75 \begin{foreignabstract}
76
77 In this work, we present an example abstract. Lorem ipsum dolor sit amet,
78 consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et
79 dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation
80 ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor
81 in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla
82 pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui
83 officia deserunt mollit anim id est laborum. Sed ut perspiciatis unde omnis
84 iste natus error sit voluptatem accusantium doloremque laudantium, totam rem
85 aperiam, eaque ipsa quae ab illo inventore veritatis et quasi architecto beatae
86 vitae dicta sunt explicabo, nemo enim ipsam voluptatem quia voluptas sit
87 aspernatur aut odit aut fugit.
88
89 \end{foreignabstract}
90
91 % Ordem pre-textual (NBR 14724/6027, R55): listas (ilustracoes, tabelas,
92 % abreviaturas e siglas, simbolos) ANTES do sumario; sumario por ultimo.
93 \listoffigures
94 \listoftables
95 \listofquadros
96 \listofprogramas
97 \listofalgorithms
98 \printloabbreviations
99 \printlosymbols
100 \tableofcontents

```

101

102 `\mainmatter`

103 `\chapter{Introdução}`

104

105 Este é um documento exemplo para o uso da classe CoppeTeX, destinado a ajudar os alunos da

106

107 A classe `\verb|coppe|` foi criada originalmente por Vicente Helano e George Ainsworth, poré

108 Em 2026, Geraldo Xexéo, com forte ajuda das ferramentas Claude 4.7 Opus e Max e 4.8 Opus M

109

110 Provavelmente Geraldo Xexéo, professor do Programa de Engenharia de Sistemas e Computação

111

112 A versão mais atual dessa classe é mantida no GitHub, no repositório `\url{https://github.c`

113

114 Esse documento segue a norma de formatação de teses e dissertações da COPPE. Ele também po

115

116 Esse documento é usado como exemplo de coisas que podem ser feitas. Ele está configurado p

117

118 É importante de notar que essa classe não foi construída sobre a classe `\LaTeX \` para a A

119

120 Essa classe implementa citações ABNT modernas e vários tipos de entrada descritos na norma

121

122 Este documento não substitui, mas complementa, o documento que descreve a classe.

123

124

125

126 `\chapter{Configurações Iniciais}`

127

128 A primeira coisa a fazer é escolher o tipo de documento. Isso é feito como uma opção no c

129

130 Como pode ser visto nesse documento, muita coisa pode ser configurada, o que gerará o tra

131

132 Recomendo ler o documento “‘The `\verb*|coppe|` document class’” para entender melhor todas

133

134 `\section{Linguagem principal do texto}`

135

136 Essa classe considera que o texto principal está em português e algumas partes específicas

137 `\begin{itemize}`

138 `\item` A opção `\verb|english|` deve ser usada no comando `\verb|\documentclass|`.

139 `\item` A bibliografia segue automaticamente o idioma do Babel (português ou inglês); não é p

140 `\end{itemize}`

141

142 A variação de linguagem, em inglês ou português apenas, já é suportada pela classe `\CoppeT`

143

144

145 `\section{Por que usar o \LaTeX}`

146

147 Há uma grande discussão entre usuário de Word e `\LaTeX`, principalmente, quanto ao uso dess

148

149 Nós escolhemos o `\LaTeX` por alguns motivos: grande facilidade de seguir um estilo sem se p

150

151 As principais desvantagens são: idiossincrasias que podem gastar tempo, pouco controle sob

152

153 `\section{Como e onde usar o \LaTeX}`

154

155 Existem muitos tutoriais de `\LaTeX`, mas basicamente, em 2024, ele é usado em dois ambientes
156 `\begin{enumerate}`
157 `\item` Na sua máquina, instalando uma versão completa como o `\hologo{MiKTeX}`, típico do Windows
158 `\item` Usar um ambiente na rede, como o Overleaf.
159 `\end{enumerate}`
160
161 Em todo caso, recomendo fortemente que, ao mesmo tempo, mantenha versões no Git e faça o backup
162
163
164 `\chapter{Algumas Regras da COPPE}`
165
166 Todas abreviaturas e símbolos devem ser definidas antes de utilizadas. Isso é facilmente feito
167
168 É imprescindível definir os símbolos, tal como o
169 conjunto dos números reais $\mathbb{R}$ e o conjunto vazio $\emptyset$.
170 `\syml{\mathbb{R}}{Conjunto dos números reais}`
171 `\syml{\emptyset}{Conjunto vazio}`. Usamos esse exemplo aqui justamente para mostrar como
172
173 Para as listas de abreviaturas e símbolos funcionarem no Overleaf é necessário rodar o `\verb|makeindex|`
174
175 Como as listas de símbolos e de abreviaturas usamos o mesmo comando usado para criar índices
176
177 `\section{Citações}`
178 `\label{sec:citacoes}`
179
180 Citações curtas podem ser feitas `\quote{o comando quote}` ou direto com “duas crases e dois pontos”
181
182 `\begin{longquote}`
183 Um exemplo de citação longa nas regras da ABNT (4cm de recuo e fonte menor)
184 feita com o ambiente `\verb|longquote|` The primary objective of this
185 investigation was to determine the feasibility of detecting corrosion in
186 aluminum Naval aircraft components with neutron radiographic interrogation
187 and the use of standard corrosion penetrameters. Secondary objectives
188 included the determination of the effect of object thickness on image quality,
189 the defining of minimum levels of detectability and a preliminary investigation
190 of a means whereby the degree of corrosion could be quantified with neutron
191 radiographic data. `\cite{article-example}`
192 `\end{longquote}`
193
194 Citações devem apontar as referências. Para isso, está disponível o ótimo pacote `\verb|natbib|`
195
196 Em todo caso, `\textbf{deve}` se tomar enorme atenção com as citações, para evitar ocorrer erros
197
198 Quando se cita uma obra por meio de outra (`\textit{apud}`), pela ABNT entra na lista de referências
199
200 `\chapter{Floats}`
201
202 Grande parte dos problemas de iniciantes, e veteranos, em `\LaTeX` é da localização dos `\textit{floats}`
203
204 A regra geral de posicionamento é que uma figura ou quadro só pode aparecer a partir da mesma página
205
206 `\textbf{Pela norma atual (manual2024), a legenda}` `\verb|caption|` `\textbf{de toda ilustração}`
207
208 Quadros são ilustrações cujos dados são delimitados por linhas em `\emph{todas}` as bordas (ao menos as

```

209
210
211 \section{Tabelas e Figuras Padrão}
212
213 Vamos ver uma tabela padrão, como a \autoref{tab:exemplo_numeros}.
214
215 \begin{table}[ht]
216 \centering % Centraliza a tabela
217 \caption{Exemplo de Tabela de Números}
218 \label{tab:exemplo_numeros}
219 \begin{tabular}{ccc} % Define a quantidade de colunas
220 \hline % Linha superior
221 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ % Cabeçalhos
222 \hline % Linha média
223 1 & 2 & 3 \\ % Primeira linha de dados
224 \hline
225 4 & 5 & 6 \\ % Segunda linha de dados
226 \hline
227 7 & 8 & 9 \\ % Terceira linha de dados
228 \hline
229 10 & 11 & 12 \\ % Quarta linha de dados
230 \hline % Linha inferior
231 \end{tabular}
232 \fonte{Elabora\c{c}\~ao pr\'opria.} % fonte obrigatoria, abaixo do float
233 \end{table}
234
235
236
237 Já a \autoref{fig:exemplo_figura} é uma figura padrão, com controle da largura.
238
239 \begin{figure}[ht]
240 \centering % Centraliza a figura
241 \caption{Exemplo de Figura com Legenda Acima} % Legenda em cima (norma atual)
242 \label{fig:exemplo_figura} % Etiqueta para referência cruzada
243 \includegraphics[width=0.5\textwidth]{coppe-logo.pdf} % Inclui a imagem com metade da largura
244 \source{COPPE/UFRJ.} % fonte obrigatoria, abaixo da figura
245 \end{figure}
246
247 Quadros têm a mesma regra (legenda acima, fonte abaixo), mas com linhas em todas as bordas.
248
249 \begin{quadro}[ht]
250 \centering
251 \caption{Exemplo de Quadro}
252 \label{qua:exemplo}
253 \begin{tabular}{|l|l|}
254 \hline
255 \textbf{Ilustração} & \textbf{Delimitação} \\
256 \hline
257 Tabela & Linhas apenas no topo e na base. \\
258 \hline
259 Quadro & Linhas em todas as bordas. \\
260 \hline
261 \end{tabular}
262 \fonte{Elabora\c{c}\~ao pr\'opria.}

```

```

263 \end{quadro}
264
265
266
267
268
269 \section{Tabelas mais elegantes}
270
271 Atualmente a tendência é usar tabelas mais leves, como \autoref{tab:exemplo_numerosbom}. Iss
272
273 \begin{table}[ht]
274 \centering % Centraliza a tabela
275 \caption{Exemplo de Tabela de Números mais elegantes}
276 \label{tab:exemplo_numerosbom}
277 \begin{tabular}{ccc} % Define a quantidade de colunas
278 \toprule % Linha superior
279 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \ % Cabeçalhos
280 \midrule % Linha média
281 1 & 2 & 3 \\ \ % Primeira linha de dados
282 4 & 5 & 6 \\ \ % Segunda linha de dados
283 7 & 8 & 9 \\ \ % Terceira linha de dados
284 10 & 11 & 12 \\ \ % Quarta linha de dados
285 \bottomrule % Linha inferior
286 \end{tabular}
287 \fonte{Elabora\c{c}\~ao pr\'opria.}
288 \end{table}
289
290 \section{Tabelas Longas ou Largas}
291
292 Se sua tabela é muito longa ou larga, existem várias opções.
293 \begin{itemize}
294 \item alterar o tamanho da letra
295 \item Usar o longtable
296 \item rodar a tabela, fazendo ela em \textit{landscape}
297 \item fazer a tabela dentro de um minibox
298 \end{itemize}
299
300
301 \subsection{Tabelas largas demais}
302
303 É comum em teses que as tabelas sejam largas demais. Há várias formas de resolver isso.
304
305 A \autoref{tab:tabela_largafns} é larga demais, e nela isso é resolvido diminuindo a fonte p
306
307 \begin{table}[ht]
308 \centering % Centraliza a tabela
309 \caption{Exemplo de Tabela Larga com Fonte Menor}
310 \label{tab:tabela_largafns}
311 \footnotesize % Aplica uma fonte menor para a tabela
312 \begin{tabular}{ccccccc} % Aumente o número de colunas conforme necessário
313 \toprule
314 \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} & \textbf{Coluna 4} & \textbf{Colu
315 \midrule
316 Dado 1.1 & Dado 1.2 & Dado 1.3 & Dado 1.4 & Dado 1.5 & Dado 1.6 & Dado 1.7 & Dado 1.8 \\

```

```

317 Dado 2.1 & Dado 2.2 & Dado 2.3 & Dado 2.4 & Dado 2.5 & Dado 2.6 & Dado 2.7 & Dado 2.8 \\
318 Dado 3.1 & Dado 3.2 & Dado 3.3 & Dado 3.4 & Dado 3.5 & Dado 3.6 & Dado 3.7 & Dado 3.8 \\
319 \bottomrule
320 \end{tabular}
321 \fonte{Elabora\c{c}\~ao pr\'opria.}
322 \end{table}
323
324 O comando \verb|\resizebox{width}{height}{content}| permite ajustar o tamanho de qualquer co
325
326 \begin{table}[ht]
327 \centering
328 \caption{Exemplo de Tabela Redimensionada}
329 \label{tab:examplerb}
330 \resizebox{\textwidth}{!}{%
331 \begin{tabular}{llll}
332 \toprule
333 Coluna 1 & Coluna 2 & Coluna 3 & Coluna 4 \\
334 \midrule
335 Dados 1 & Dados 2 & Dados 3 & Dados 4 \\
336 Dados 5 & Dados 6 & Dados 7 & Dados 8 \\
337 \bottomrule
338 \end{tabular}%
339 }
340 \fonte{Elabora\c{c}\~ao pr\'opria.}
341 \end{table}
342
343
344 Para rodar uma tabela muito larga em 90 graus no LaTeX, voc\^e pode usar o pacote \verb*|rotat
345
346 Aqui est\^a um exemplo de como usar o ambiente \verb*|sidewaystable| para girar uma tabela. Pr
347
348 \begin{sidewaystable}
349 \centering
350 \caption{Sua Legenda Aqui}
351 \label{tab:sua_tabela}
352 \begin{tabular}{llll}
353 \toprule
354 Coluna 1 & Coluna 2 & Coluna 3 \\
355 \midrule
356 Item 1 & Item 2 & Item 3 \\
357 Item 4 & Item 5 & Item 6 \\
358 \bottomrule
359 \end{tabular}
360 \fonte{Elabora\c{c}\~ao pr\'opria.}
361 \end{sidewaystable}
362
363 Se a tabela for muito longa, o ambiente \verb|longtable| \^e o ideal. Ele fornece comandos par
364
365 % Exemplo de tabela longa que se estende por v\^arias p\^aginas
366 \begin{longtable}{|c|c|c|}
367 % primeiro cabe\c{c}alho (\^e o caption)
368 \caption{Exemplo de Tabela Longa}\label{tab:longa} \\
369 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \\ \hline
370 \endfirsthead

```

```

371 % cabeçalho normal
372 \multicolumn{3}{c}%
373 {\table\ \thetable} -- continuação da página anterior} \
374 \hline \textbf{Coluna 1} & \textbf{Coluna 2} & \textbf{Coluna 3} \ \hline
375 \endhead
376 % pé normal
377 \hline \multicolumn{3}{r|}{Continua na próxima página} \ \hline
378 \endfoot
379 \hline
380 % último pé
381 \multicolumn{3}{r|}{Continua na próxima página}} \
382 \hline \hline
383 \endlastfoot
384
385 % Conteúdo da tabela
386 1 & 2 & 3 \
387 4 & 5 & 6 \
388 1 & 2 & 3 \
389 4 & 5 & 6 \
390 1 & 2 & 3 \
391 4 & 5 & 6 \
392 1 & 2 & 3 \
393 4 & 5 & 6 \
394 1 & 2 & 3 \
395 4 & 5 & 6 \
396 1 & 2 & 3 \
397 4 & 5 & 6 \
398 1 & 2 & 3 \
399 4 & 5 & 6 \
400 1 & 2 & 3 \
401 4 & 5 & 6 \
402 1 & 2 & 3 \
403 1 & 2 & 3 \
404 4 & 5 & 6 \
405 1 & 2 & 3 \
406 4 & 5 & 6 \
407 1 & 2 & 3 \
408 4 & 5 & 6 \
409 1 & 2 & 3 \
410 4 & 5 & 6 \
411 1 & 2 & 3 \
412 4 & 5 & 6 \
413 1 & 2 & 3 \
414 4 & 5 & 6 \
415 1 & 2 & 3 \
416 4 & 5 & 6 \
417 1 & 2 & 3 \
418 4 & 5 & 6 \
419 1 & 2 & 3 \
420 4 & 5 & 6 \
421 1 & 2 & 3 \
422 4 & 5 & 6 \
423 1 & 2 & 3 \
424 4 & 5 & 6 \

```

425 1 & 2 & 3 \\
 426 4 & 5 & 6 \\\ 1 & 2 & 3 \\
 427 4 & 5 & 6 \\
 428 1 & 2 & 3 \\
 429 4 & 5 & 6 \\
 430 1 & 2 & 3 \\
 431 1 & 2 & 3 \\
 432 4 & 5 & 6 \\
 433 1 & 2 & 3 \\
 434 4 & 5 & 6 \\
 435 1 & 2 & 3 \\
 436 4 & 5 & 6 \\
 437 1 & 2 & 3 \\
 438 4 & 5 & 6 \\
 439 1 & 2 & 3 \\
 440 4 & 5 & 6 \\
 441 1 & 2 & 3 \\
 442 4 & 5 & 6 \\
 443 1 & 2 & 3 \\
 444 4 & 5 & 6 \\
 445 1 & 2 & 3 \\
 446 4 & 5 & 6 \\
 447 1 & 2 & 3 \\
 448 4 & 5 & 6 \\
 449 1 & 2 & 3 \\
 450 4 & 5 & 6 \\
 451 1 & 2 & 3 \\
 452 4 & 5 & 6 \\
 453 1 & 2 & 3 \\
 454 4 & 5 & 6 \\
 455 1 & 2 & 3 \\
 456 4 & 5 & 6 \\
 457 1 & 2 & 3 \\
 458 4 & 5 & 6 \\
 459 1 & 2 & 3 \\
 460 4 & 5 & 6 \\
 461 1 & 2 & 3 \\
 462 4 & 5 & 6 \\
 463 1 & 2 & 3 \\
 464 4 & 5 & 6 \\
 465 1 & 2 & 3 \\
 466 4 & 5 & 6 \\
 467 1 & 2 & 3 \\
 468 4 & 5 & 6 \\
 469 1 & 2 & 3 \\
 470 4 & 5 & 6 \\
 471 1 & 2 & 3 \\
 472 4 & 5 & 6 \\
 473 1 & 2 & 3 \\
 474 4 & 5 & 6 \\
 475 1 & 2 & 3 \\
 476 4 & 5 & 6 \\
 477 1 & 2 & 3 \\
 478 4 & 5 & 6 \\

```

479 1 & 2 & 3 \\
480 4 & 5 & 6 \\
481 1 & 2 & 3 \\
482 4 & 5 & 6 \\
483 1 & 2 & 3 \\
484 4 & 5 & 6 \\
485 1 & 2 & 3 \\
486 4 & 5 & 6 \\
487 1 & 2 & 3 \\
488 4 & 5 & 6 \\
489 1 & 2 & 3 \\
490 4 & 5 & 6 \\
491 1 & 2 & 3 \\
492 4 & 5 & 6 \\
493
494 % Repetir linhas semelhantes conforme necessário para estender a tabela por 3 páginas
495 \end{longtable}
496 \fonte{Elabora\c{c}\~ao pr\'opria.}% a longtable não é float: a fonte vem após o ambiente
497
498 \section{Subfiguras (subcaption)}
499
500 Para colocar várias imagens (ou tabelas) lado a lado, cada uma com sua própria legenda, use
501
502 \begin{figure}[ht]
503   \centering
504   \begin{subfigure}[b]{0.42\textwidth}
505     \centering\includegraphics[width=\linewidth]{example-image-a}
506     \caption{Primeira vista}\label{fig:sub-a}
507   \end{subfigure}
508   \hfill
509   \begin{subfigure}[b]{0.42\textwidth}
510     \centering\includegraphics[width=\linewidth]{example-image-b}
511     \caption{Segunda vista}\label{fig:sub-b}
512   \end{subfigure}
513   \caption{Figura com duas subfiguras}
514   \label{fig:subs}
515   \fonte{Elabora\c{c}\~ao pr\'opria.}
516 \end{figure}
517
518 Referencie a figura inteira com \verb|\autoref{fig:subs}| e cada parte com \verb|\autoref{fig:sub-a}|
519
520 \section{Imagens (graphicx)}
521
522 A classe já carrega o \verb|graphicx|. Insira imagens com \verb|\includegraphics|; as opções
523 \begin{itemize}
524   \item \verb|[width=0.8\textwidth]| ou \verb|[scale=0.5]| --- tamanho;
525   \item \verb|[angle=90]| --- rotação;
526   \item \verb|[trim=l b r t, clip]| --- recorte (corta as bordas \emph{esquerda, baixo, direita, topo});
527 \end{itemize}
528 Formatos aceitos com \hologo{pdfLaTeX}/\hologo{LuaLaTeX}: \texttt{pdf}, \texttt{png}, \texttt{eps}
529
530 \section{Introdução ao tikz}
531
532 0 \verb|tikz| (disponível também via \verb|tcolorbox|) desenha gráficos vetoriais no próprio

```

```

533
534 \begin{figure}[ht]
535   \centering
536   \begin{tikzpicture}
537     \node[draw, rounded corners] (a) at (0,0) {Início};
538     \node[draw, rounded corners] (b) at (4,0) {Fim};
539     \draw[->, thick] (a) -- (b) node[midway, above] {processo};
540   \end{tikzpicture}
541   \caption{Diagrama simples em tikz}
542   \label{fig:tikz}
543   \fonte{Elabora\c{c}\~ao pr\'opria.}
544 \end{figure}
545
546 Para gráficos de dados (curvas, barras), veja o \verb|pgfplots|, construído sobre o \verb|tikz|
547
548 \section{Float ou direto no texto?}
549
550 Colocar a tabela ou imagem dentro de \verb|figure|/\verb|table| (um \emph{float}) deixa o \verb|float|
551
552 \chapter{Using BibLaTeX}
553 \label{cap:biblatex}
554
555 Esta classe gera as referências com BibLaTeX e o \emph{backend} \texttt{biber}, no padrão
556
557 \textbf{É} obrigatório compilar com \texttt{biber} (e não com \hologo{BibTeX}). A sequência
558
559 Os comandos do \texttt{natbib} (\verb|\citet|, \verb|\citep|) continuam disponíveis porque
560
561 \begin{table}[h]
562 \caption{Exemplos de cita\c{c}\~oes utilizando o comando padr\~ao}
563 \texttt{\textbackslash cite} do \LaTeX\ e
564 o comando \texttt{\textbackslash citet},
565 disponível pela opção \texttt{natbib} do BibLaTeX.
566 \label{tab:citation}
567 \centering
568 {\footnotesize
569 \begin{tabular}{|c|c|c|}
570 \hline
571 Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
572 \hline
573 Livro & \cite{book-example} & \citet{book-example}\\
574 Artigo & \cite{article-example} & \citet{article-example}\\
575 Relatório & \cite{techreport-example} & \citet{techreport-example}\\
576 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
577 Anais de Congresso & \cite{inproceedings-example} & \citet{inproceedings-example}\\
578 Séries & \cite{incollection-example} & \citet{incollection-example}\\
579 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
580 Dissertação de mestrado & \cite{mastersthesis-example} & \citet{mastersthesis-example}\\
581 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
582 \hline
583 \end{tabular}}
584 \end{table}
585 \fonte{Elabora\c{c}\~ao pr\'opria.}
586

```

```

587 \end{table}
588
589 \begin{table}[h]
590 \caption{Exemplos de cita{\c c}\~oes utilizando o comando padr\~ao
591 \texttt{\textbackslash cite} do \LaTeX\ e
592 o comando \texttt{\textbackslash citet},
593 disponível pela opção \texttt{\natbib} do BibLaTeX. Além disso, usando o booktabs.}
594 \label{tab:citation1}
595 \centering
596 {\footnotesize
597 \begin{tabular}{ccc}
598 \toprule
599 Tipo da Publicação & \verb|\cite| & \verb|\citet|\\
600 \midrule
601 Livro & \cite{book-example} & \citet{book-example}\\
602 Artigo & \cite{article-example} & \citet{article-example}\\
603 Relatório & \cite{techreport-example} & \citet{techreport-example}\\
604 Relatório & \cite{techreport-exampleIn} & \citet{techreport-exampleIn}\\
605 Anais de Congresso & \cite{inproceedings-example} &
606 \citet{inproceedings-example}\\
607 Séries & \cite{incollection-example} & \citet{incollection-example}\\
608 Em Livro & \cite{inbook-example} & \citet{inbook-example}\\
609 Dissertação de mestrado & \cite{mastersthesis-example} &
610 \citet{mastersthesis-example}\\
611 Tese de doutorado & \cite{phdthesis-example} & \citet{phdthesis-example}\\
612 \bottomrule
613 \end{tabular}}
614 \fonte{Elabora\c c}\~ao pr\'opria.}
615 \end{table}
616
617 \section{Comandos de citação}
618
619 Os principais comandos são: \verb|\cite| (referência simples), \verb|\citet| (autor por ex
620
621 \subsection{Parâmetro opcional: página e “tradução nossa”}
622
623 Todos os comandos aceitam um parâmetro opcional para a localização (página, seção) e obser
624
625 \subsection{Citação de citação (\emph{apud})}
626
627 Para citar uma obra através de outra, use \verb|\citetapud| e \verb|\citepapud| (demonstra
628
629 \subsection{Múltiplas bibliografias}
630
631 Para separar as referências (por tipo, capítulo ou tema), use o ambiente \verb|refsection|
632
633 \section{Tipos de entrada bibliográfica}
634
635 Cada tipo de documento da ABNT corresponde a um tipo de entrada do BibLaTeX, com um sinôni
636
637 \subsection{Tipos acadêmicos principais}
638 \begin{itemize}
639 \item Livro / monografia --- \verb|\@book| (\verb|\@livro|)
640 \item Artigo de periódico --- \verb|\@article| (\verb|\@artigo|); matéria de jornal --- \verb|\@journal|

```

```

641 \item Capítulo de livro --- \verb|@incollection| (\verb|@partedelivro|), \verb|@inbook|
642 \item Evento no todo (anais) --- \verb|@proceedings| (\verb|@anais|); trabalho em evento
643 \item Tese/dissertação --- \verb|@thesis| (\verb|@tese|, \verb|@dissertacao|); com tipo
644 \item Relatório --- \verb|@report| (\verb|@relatorio|, \verb|@relatoriotecnico|); período
645 \item Manual --- \verb|@manual|; diversos --- \verb|@misc| (\verb|@diverso|); página/sit
646 \end{itemize}
647
648 \subsection{Tipos especiais (ABNT §4.2.5--4.2.14)}
649 \begin{itemize}
650 \item Patente --- \verb|@patent| (\verb|@patente|)
651 \item Legislação / ato normativo --- \verb|@legislation| (\verb|@legislacao|, \verb|@ato
652 \item Norma técnica --- \verb|@standard| (\verb|@norma|); partitura --- \verb|@music| (\
653 \item Documento cartográfico (mapa) --- \verb|@map| (\verb|@mapa|, \verb|@cartografico|)
654 \item Audiovisual --- \verb|@video| (\verb|@audiovisual|); filme --- \verb|@movie| (\verb
655 \end{itemize}
656
657 \subsection{Jogos e TV (tipos próprios da CoppeTeX)}
658 \begin{itemize}
659 \item Jogo --- \verb|@game| (\verb|@jogo|); de tabuleiro --- \verb|@boardgame| (\verb|@j
660 \item Programa de TV / série --- \verb|@tvshow| (\verb|@programadetv|, \verb|@serietv|)
661 \end{itemize}
662
663 Os campos também aceitam sinônimos em português (sem acentos): \verb|autor|, \verb|titulo|
664
665 \chapter{Alguns outros exemplo úteis}
666
667 \begin{tcolorbox}[title=Meu Textbox]
668 Este é o conteúdo do meu textbox. Você pode adicionar qualquer texto aqui, bem como incluir
669 \end{tcolorbox}
670
671 \begin{tcolorbox}
672 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
673 \end{tcolorbox}
674
675 \begin{figure}[ht]
676 \centering
677 \begin{tikzpicture}
678 \node[anchor=south west,inner sep=0] (image) at (0,0) {\includegraphics[width=0.5\te
679 \begin{scope}[x={({image.south east}}),y={({image.north west}})]
680 % Definindo o textbox dentro da figura
681 \node[anchor=north west, text width=0.3\textwidth, fill=white, opacity=0.7, text
682 \begin{tcolorbox}[colback=red!5!white,colframe=red!75!black,title=Textbox de
683 Este textbox fala sobre como inserir um textbox dentro de uma figura usa
684 \end{tcolorbox}
685 };
686 \end{scope}
687 \end{tikzpicture}
688 \caption{Figura com Textbox}
689 \label{fig:figura_com_textbox1}
690 \fonte{Elabora\c{c}\~ao pr\'opria.}
691 \end{figure}
692
693
694 \begin{figure}[ht]

```

```

695     \centering
696 \begin{tcolorbox}
697 Este é o conteúdo do meu textbox sem título. Você pode adicionar qualquer texto aqui, bem co
698 \end{tcolorbox}
699     \caption{Figura com Textbox simples}
700     \label{fig:figura_com_textbox}
701     \fonte{Elabora\c{c}\~ao pr\'opria.}
702 \end{figure}
703
704 \section{Listagens de código e algoritmos}
705
706 O CoppeTeX traz embutido o suporte a listagens de programas (pacote \verb|listings|) e a alg
707
708 \begin{python}[caption={Exemplo de programa em Python},label=prog:exemplo]
709 def saudacao(nome):
710     # imprime uma saudação acentuada
711     print(f"Olá, {nome}!")
712
713 saudacao("COPPE")
714 \end{python}
715 \source{Elaboração própria.}
716
717 Para inserir código no meio do texto use \verb|\pythoninline|, como em \pythoninline{saudaca
718
719 \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
720 Olá, COPPE!
721 \end{Verbatim}
722
723 Há estilos prontos também para \verb|java|, \verb|xml|, \verb|html| e \verb|prolog|, comando
724
725 \begin{algorithm}[H]
726 \caption{Soma dos elementos de um vetor}
727 \label{alg:soma}
728 \Dados{vetor  $v$  com  $n$  elementos}
729 \Resultado{soma  $s$  dos elementos de  $v$ }
730  $s \leftarrow 0$ ;
731 \Para{ $i \leftarrow 1$  \Ate{ }  $n$ }{
732      $s \leftarrow s + v[i]$ ;
733 }
734 \Retorna  $s$ ;
735 \end{algorithm}
736
737 \chapter{Escrevendo matemática}
738
739 A classe carrega \texttt{amsmath} em qualquer motor. Sob \hologo{pdfLaTeX} ela também carr
740
741 \section{Alinhamento (align)}
742 O ambiente \verb|align| alinha equações pelo \verb|&| e numera cada linha (\verb|\\| quebr
743 \begin{align}
744     f(x) &= (x+1)^2 \\
745         &= x^2 + 2x + 1 .
746 \end{align}
747
748 \section{Matrizes (matrix)}

```

```

749 0 \texttt{amsmath} traz \verb|pmatrix|, \verb|bmatrix|, \verb|vmatrix|, entre outros:
750 \begin{equation}
751   A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}
752   \quad
753   \det A = \begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix}
754 \end{equation}
755
756 \section{Subequações (subequations)}
757 Para numerar um grupo de equações sob o mesmo número, com letras (a), (b):
758 \begin{subequations}
759 \begin{align}
760   a &= b + c , \\
761   d &= e - f .
762 \end{align}
763 \end{subequations}
764
765 \section{Estruturas alinhadas (array)}
766 0 \verb|array| monta estruturas dentro do modo matemático, como definições por partes:
767 \begin{equation}
768   |x| = \left\{ \begin{array}{l}
769     x, \quad \text{se } x \geq 0, \\
770     -x, \quad \text{caso contrário.}
771   \end{array} \right.
772 \end{equation}
773
774 \section{Operadores (\textbackslash DeclareMathOperator)}
775 Para operadores com a tipografia correta (romano, espaçamento adequado), declare-os no pre
776 \begin{verbatim}
777 \usepackage{amsmath}
778 \DeclareMathOperator{\RMSE}{RMSE}
779 \DeclareMathOperator{\diag}{diag}
780 \end{verbatim}
781 Assim, \verb|$ \RMSE(y, \hat{y})| produz  $\RMSE(y, \hat{y})$ , e \verb|$ \diag(\lambda_1, \dots
782
783 \section{Unicode e LuaLaTeX (unicode-math)}
784 \label{sec:unicode-math}
785 Ao migrar para \hologo{LuaLaTeX} (ou \hologo{XeLaTeX}), \textbf{não carregue} \texttt{amss
786 \begin{verbatim}
787 \usepackage{amsmath} % a classe já carrega
788 \usepackage{unicode-math} % NÃO use \usepackage{amssymb} junto
789 \setmathfont{Latin Modern Math} % ou STIX Two Math, TeX Gyre Termes Math...
790 \end{verbatim}
791 0 código clássico continua valendo ( $\mathbb{R}$ ),  $\int_a^b f(x) dx$ ,  $\alpha$ ) e você ai
792
793 \chapter{Referências cruzadas, notas e índices}
794 \label{cap:refs}
795
796 Este capítulo reúne os recursos para \emph{apontar} para coisas: rótulos, notas de rodapé,
797
798 \section{Rótulos e referências cruzadas}
799
800 Marque um elemento com \verb|\label{tipo:nome}| e aponte para ele com \verb|\ref| (só o nú
801
802 \section{Notas de rodapé}

```

```

803
804 Use \verb|\footnote{texto}|\footnote{Como esta nota, que a classe imprime em espaçamento s
805
806 \section{hyperref e marcadores}
807
808 A classe carrega o \verb|hyperref|: o sumário, as citações e os \verb|\ref|/\verb|\autoref|
809
810 \section{URLs}
811
812 Para endereços, use \verb|\url{https://exemplo.org}|: o \verb|hyperref| cria o link e queb
813
814 \section{Índice remissivo}
815
816 Marque os termos com \verb|\index{termo}| e imprima o índice com \verb|\printindex| (carre
817
818 \section{Traduções (NBR 10520)}
819
820 Ao citar um trecho que você mesmo traduziu, coloque a tradução no corpo do texto e escreva
821
822 \chapter{Truques de \LaTeX}
823
824 \section{Comentários mágicos (\texttt{\% !TeX})}
825 Linhas \verb|\% !TeX| no topo do arquivo dizem ao editor (\hologo{TeX}studio, Overleaf, \ho
826 \begin{verbatim}
827 % !TeX program = lualatex
828 % !TeX encoding = UTF-8
829 % !TeX spellcheck = pt_BR
830 % !BIB program = biber
831 \end{verbatim}
832 Com isso o documento compila igual em qualquer ambiente. O \texttt{latexmkrc} cumpre papel
833
834 \section{Espaço engolido}
835 Um comando sem argumento engole o espaço seguinte: \verb|\LaTeX é bom| sai como ‘\LaTeX é
836
837 \section{Escopo (grupos)}
838 Chaves \verb|\{...}| (ou \verb|\begingroup\...\endgroup|) criam um escopo: mudanças
839
840 \section{Descarregar figuras: \texttt{\textbackslash clearpage} × \texttt{\textbackslash m
841 \verb|\newpage| apenas quebra a página e leva os floats pendentes adiante, acumulando-os.
842
843 \section{\texttt{latexmkrc}, perl e Overleaf}
844 O \verb|latexmk| automatiza a compilação (roda \LaTeX, \verb|biber| e \verb|makeindex| qua
845
846 \chapter{Método Proposto}
847 \chapter{Resultados e Discussões}
848
849 \section{Algumas Demonstrações}
850
851 A Lista de Símbolos precisa usar comandos específicos. Aqui vamos usar os símbolos $\alpha$
852 \syml[beta]{Beta}{A palavra Beta more e corrigida}
853 \syml[zzbeta]{\beta}{A letra $\beta$ corrigida}
854 \syml[beta]{A palavra beta}
855 \syml[alpha]{A palavra alpha}
856 \syml[alpha]{Alpha}{A palavra Alpha}

```

```

857 \syml[zzalpha]{\alpha}{A letra \alpha corrigida}
858 \syml[marco]{Marco}{A palavra Marco corrigida}
859
860 A Lista de Abreviações segue, a partir de 2024, a mesma regra, e aqui seguem alguns exemplos
861 \abbrev{GoT}{Game of Thrones}
862 \abbrev[GOT]{GoT}{Game of Thrones ordenado como GOT}
863 \abbrev[iot]{IoT}{IoT ordenado como iot}
864 \abbrev[IoT]{IoT}{IoT ordenado como IoT}
865 \abbrev[IOT]{IoT}{IoT ordenado como IOT}
866 \abbrev{IoT}{IoT com ordenação default}
867 \abbrev[ITU]{ITU}{ITU mesmo}
868
869 \newsigla{ufrj}{UFRJ}{Universidade Federal do Rio de Janeiro}
870 \newsigla{cpgp}{CPGP}{Comissão de Pós-Graduação e Pesquisa de Engenharia}
871
872 As siglas (manual2024, seção 2.8) são expandidas na primeira ocorrência e registradas automaticamente
873
874
875
876
877 \chapter{Conclusões}
878
879 \backmatter
880
881 % Tipos exóticos da ABNT (NBR 6023): patente, legislação, norma, mapa,
882 % partitura --- declarados via sinônimos em português em example.bib.
883 \nocite{patente-example,legislacao-example,norma-example,mapa-example,partitura-example}
884 \nocite{jurisprudencia-example}
885 \nocite{videogame-example,filme-example,semautor-example,semautorart-example}
886 \nocite{jornalcaderno-example,jornal-example}
887 \printbibliography
888
889
890
891 \appendix
892
893 \chapter{Um apêndice}
894
895 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização de apêndices
896
897 Apêndice: O apêndice é um texto ou documento elaborado pelo autor do trabalho com o objetivo de complementar
898
899
900 \chapter{Outro apêndice}
901
902 \annex
903
904
905 \chapter{Um Anexo}
906 Segundo a norma da ABNT (Associação Brasileira de Normas Técnicas), a definição e utilização de anexos
907
908
909
910 Anexo: O anexo, por sua vez, consiste em um texto ou documento não elaborado pelo autor, que

```

```

911
912 No modelo \CoppeTeX os anexos devem obrigatoriedade vir depois dos apêndices e usam o comand
913
914
915 \chapter{Outro Anexo}
916
917
918 \coppetexfinalpage% colofão institucional (página par, face do contracapa)
919
920 \end{document}
921
922 </example>

```

10 Implementation

10.1 The ‘coppe.cls’ file

```

923 <*class>
924 \def\filename{coppe.dtx}
925 \def\fileversion{v4.0}
926 \def\filedate{2026/05/28}
927 \NeedsTeXFormat{LaTeX2e}[1995/12/01]
928 \ProvidesClass{coppe}[\filedate\ \fileversion\ COPPE Dissertations and Thesis]
929 % Save the class version and date into private macros immediately, before any
930 % required package can redefine the very common \fileversion / \filedate
931 % global names (LaTeX has no convention reserving them for the host class).
932 % \coppe@version / \coppe@date are what user-facing macros (e.g.
933 % \coppetexfinalpage) should reference.
934 \let\coppe@version\fileversion
935 \let\coppe@date\filedate
936 % Document date stored in private slots so \date can set them without ever
937 % touching TeX's \month/\year primitives (which would break \today). They
938 % default to the current system month/year, so a document without \date
939 % still renders sensibly on the cover and folha de rosto.
940 \edef\coppe@docmonth{\the\month}
941 \edef\coppe@docyear{\the\year}
942 \LoadClass[12pt,a4paper,twoside]{book}
943 % setspace, hyperref and the bibliography engine (biblatex+biber) are loaded at
944 % the END of the class (after every class hook is defined), in that order:
945 % setspace -> hyperref -> biblatex. hyperref must come late so all the macros
946 % it patches (TOC, refs, lists) are already defined, and biblatex must come
947 % after hyperref so its citation links and back-references work.
948 \RequirePackage{hyphenat}
949 \RequirePackage{lastpage}
950 \RequirePackage{ifthen}
951 \RequirePackage{graphicx}
952 \RequirePackage{tabularx}
953 \RequirePackage{booktabs} % regras de tabela profissionais (toprule/midrule/bottomrule)
954 \RequirePackage{longtable} % tabelas que quebram entre páginas
955 \RequirePackage{etoolbox}
956 \RequirePackage{ltxcmds}
957 \RequirePackage{hologo} % TeX-family logos: \hologo{TeX}, \hologo{LaTeX}, ...
958 % Render \TeX, \LaTeX and friends via hologo (\CoppeTeX is a custom logo, kept).
959 \def\TeX{\hologo{TeX}}

```

```

960 \def\LaTeX{\hologo{LaTeX}}
961 \def\LaTeXe{\hologo{LaTeXe}}
962 \def\BibTeX{\hologo{BibTeX}}
963 \def\pdfLaTeX{\hologo{pdfLaTeX}}
964 \def\pdfTeX{\hologo{pdfTeX}}
965 \def\LuaLaTeX{\hologo{LuaLaTeX}}
966 \def\XeLaTeX{\hologo{XeLaTeX}}
967 \def\MiKTeX{\hologo{MiKTeX}}
968 % Engine-aware UTF-8 input: pdfLaTeX needs inputenc(utf8); LuaLaTeX (and XeTeX)
969 % are UTF-8 native, so inputenc must NOT be loaded there (it would clash). Source
970 % files may use accented Latin characters (áéíóú) directly. Under pdfLaTeX we also
971 % suggest LuaLaTeX, which handles Unicode/fonts better (at the cost of speed).
972 \RequirePackage{iftex}
973 \ifPDFTeX
974   \RequirePackage[utf8]{inputenc}
975   \AtEndOfClass{\ClassWarningNoLine{coppe}{%
976     Although slower, you should try compiling with LuaLaTeX}}
977 \fi
978 \RequirePackage[T1]{fontenc}
979 % Math support. amsmath is universal and works on every engine, so the class
980 % brings it in unconditionally (it must precede hyperref, which is loaded at
981 % the end of the class). amssymb is the legacy AMS symbol package, built
982 % around the 8-bit Type 1 fonts msam/msbm; on a Unicode engine (LuaLaTeX or
983 % XeLaTeX) it clashes with unicode-math, which already provides all of its
984 % symbols through Unicode math codes. So amssymb is loaded ONLY under
985 % pdfLaTeX; Unicode-engine users get a ClassWarning pointing them at
986 % unicode-math (the class does not auto-load unicode-math because choosing
987 % an OpenType math font is a typographic decision that belongs to the author).
988 \RequirePackage{amsmath}
989 \ifPDFTeX
990   \RequirePackage{amssymb}
991 \else
992   % Unicode engine: defer the check to end of document. Only warn if the
993   % user did NOT add either unicode-math (the modern OpenType-math route)
994   % or amssymb (the legacy route, which they would have loaded manually).
995   % This avoids nagging authors who already followed the documented
996   % \usepackage{unicode-math} workflow.
997   \AtEndDocument{%
998     \@ifpackageloaded{unicode-math}{\}%
999     \@ifpackageloaded{amssymb}{\}%
1000     \ClassWarningNoLine{coppe}{%
1001       You are compiling with a Unicode engine (LuaLaTeX or XeLaTeX) and\MessageBreak
1002       neither unicode-math nor amssymb is loaded. AMS-style symbols\MessageBreak
1003       (\string\mathbb, \string\hbar, \string\varnothing, ...) will be undefined.\MessageBreak
1004       Add to your preamble:\MessageBreak
1005       \space\space\string\usepackage{unicode-math}\MessageBreak
1006       \space\space\string\setmathfont{Latin Modern Math}\MessageBreak
1007       Do NOT load amssymb together with unicode-math -- it would clash\MessageBreak
1008       with the OpenType math font setup}}}%
1009 \fi
1010 % manual2024 margins: 3cm top, 2cm bottom, mirrored 3cm inner / 2cm outer for
1011 % two-sided printing -- realised as inner=outer=2cm plus a 1cm bindingoffset.
1012 % headheight=15pt is given here (not patched later) so geometry's own page
1013 % arithmetic stays self-consistent: fancyhdr needs >= 14.49998pt of header

```

```

1014 % height to host the running head, and the standard 12pt default triggers
1015 % its "headheight is too small" warning. Letting geometry know up front
1016 % keeps \topmargin a derived value -- we never touch it by hand.
1017 \RequirePackage[a4paper,twoside,bindingoffset=1cm,headheight=15pt,%
1018 top=3cm,bottom=2cm,inner=2cm,outer=2cm]{geometry}
1019 \RequirePackage{titlesec}
1020 % --- Section heading formats (manual2024 2.6 / proposta P22) -----
1021 % chapter (numbered): "N TITLE", ALL CAPS, bold, flush left (no "Capitulo N").
1022 % APENDICE/ANEXO chapters print "APENDICE A -- Title" (R75/R76); normal chapters
1023 % keep "N Title". \appendix and \annex raise the flag (see below).
1024 \newif\if@coppeappendix \@coppeappendixfalse
1025 \titleformat{\chapter}[hang]
1026 {\normalfont\Large\bfseries}
1027 {\if@coppeappendix\MakeUppercase{\appendixname}\ \thechapter\ \textendash\else\thechapter\
1028 {lex}\MakeUppercase}
1029 \titlespacing*{\chapter}{0pt}{10pt}{1.5\baselineskip}
1030 \let\coppe@orig@appendix\appendix
1031 \renewcommand\appendix{\coppe@orig@appendix\@coppeappendixtrue}
1032 % Sumario (TOC) entries for APENDICE/ANEXO chapters read "APENDICE A -- Title"
1033 % / "ANEXO A -- Title" (R75/R76), not the bare "A". The label word is chosen by
1034 % which macro is written (so it is correct at .toc-render time even though the
1035 % appendixname has changed back); the chapter letter is baked in at write time.
1036 \newif\if@coppeannex \@coppeannexfalse
1037 \newcommand\coppe@tocapp[1]{%
1038 \MakeUppercase{\coppestring{appendix}}%
1039 \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1040 \newcommand\coppe@tocanx[1]{%
1041 \MakeUppercase{\coppestring{annex}}%
1042 \nobreakspace#1\nobreakspace\textendash\nobreakspace}
1043 \def\@chapter[#1]#2{%
1044 \ifnum \c@secnumdepth >\m@ne
1045 \if@mainmatter
1046 \refstepcounter{chapter}%
1047 \typeout{\@chapapp\space\thechapter.}%
1048 \if@coppeappendix
1049 \if@coppeannex
1050 \addcontentsline{toc}{chapter}{\protect\coppe@tocanx{\thechapter}#1}%
1051 \else
1052 \addcontentsline{toc}{chapter}{\protect\coppe@tocapp{\thechapter}#1}%
1053 \fi
1054 \else
1055 \addcontentsline{toc}{chapter}{\protect\numberline{\thechapter}#1}%
1056 \fi
1057 \else
1058 \addcontentsline{toc}{chapter}{#1}%
1059 \fi
1060 \else
1061 \addcontentsline{toc}{chapter}{#1}%
1062 \fi
1063 \chaptermark{#1}%
1064 \addtocontents{lof}{\protect\addvspace{10\p@}}%
1065 \addtocontents{lot}{\protect\addvspace{10\p@}}%
1066 \if@twocolumn
1067 \@topnewpage[\@makechapterhead{#2}]%

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1068 \else
1069 \makechapterhead{#2}%
1070 \@afterheading
1071 \fi}
1072 % chapter (unnumbered): centred ALL CAPS bold (errata, lists, resumo, sumario,
1073 % referencias, glossario, ... -- unnumbered headings are centred, P29-P30).
1074 \titleformat{name=\chapter,numberless}[hang]
1075 {\normalfont\Large\bfseries\filcenter}{\Opt}{\MakeUppercase}
1076 \titlespacing*{name=\chapter,numberless}{0pt}{10pt}{1.5\baselineskip}
1077 % section: ALL CAPS, normal weight.
1078 \titleformat{section}
1079 {\normalfont\large\bfseries}{\thesection}{1ex}{\MakeUppercase}
1080 % subsection: bold (author writes Title Case).
1081 \titleformat{subsection}
1082 {\normalfont\bfseries}{\thesubsection}{1ex}{\}
1083 % subsubsection: bold italic.
1084 \titleformat{subsubsection}
1085 {\normalfont\bfseries\itshape}{\thesubsubsection}{1ex}{\}
1086 % --- Captions (manual2024 2.10-2.11 / proposta P6): smaller font, ABNT
1087 % "Figura 1 -- titulo" travessao separator (sources handled by \source). ---
1088 \RequirePackage{caption}
1089 \captionsetup{font=footnotesize,labelsep=endash}
1090 % subcaption (NÃO subfig/subfigure) for subfigures / subtables; load right after
1091 % caption so it picks up the ABNT captionsetup above.
1092 \RequirePackage{subcaption}
1093 % --- Floats: caption on TOP and a mandatory "Fonte:" line at the bottom.
1094 % manual2024 2.10-2.11 (R46/R51: identification on top; R48/R52: the source
1095 % at the bottom is mandatory). The float package's plaintop style forces the
1096 % caption above the body for figure, table and the new quadro float. -----
1097 \RequirePackage{float}
1098 \floatstyle{plaintop}
1099 \restylefloat{figure}
1100 \restylefloat{table}
1101 % New "quadro" float -- "Quadro" (pt) / "Frame" (en) (R53) -- with its own list
1102 % file (.loq). A quadro's data is bordered on all sides (drawn by the author),
1103 % unlike a tabela (rules on top and bottom only).
1104 \providecommand{\quadroname}{\coppestring{frame}}
1105 \providecommand{\listquadroname}{\coppestring{listframe}}
1106 \newfloat{quadro}{htbp}{loq}[chapter]
1107 \floatname{quadro}{\quadroname}
1108 \providecommand{\quadroautorefname}{\quadroname}
1109 % Format the Lista de Quadros entries exactly like the Lista de Figuras ones
1110 % (float's own \l@quadro lays the number/title out incorrectly).
1111 \let\l@quadro\l@figure
1112 \newcommand\listofquadros{%
1113 \chapter*{\listquadroname}%
1114 \addcontentsline{toc}{chapter}{\listquadroname}%
1115 \mkboth{\MakeUppercase\listquadroname}{\MakeUppercase\listquadroname}%
1116 \@starttoc{loq}}
1117 % \source / \fonte: the illustration source (ABNT-mandatory). Typeset below the
1118 % float as a smaller, single-spaced, centred line; it does NOT advance any
1119 % caption counter. \cpsource/\cpfonte are clash-safe aliases (the rename rule);
1120 % the label is "Fonte" (pt) / "Source" (en).
1121 \newcommand{\cpsourcenam}{\coppestring{source}}

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1122 \newcommand{\cpsource}[1]{%
1123   \par\addvspace{\smallskipamount}%
1124   {\footnotesize\singlespacing\centering\cpsourcename: #1\par}}
1125 \let\cpfonte\cpsource
1126 \@ifundefined{source}{\let\source\cpsource}{}%
1127 \@ifundefined{fonte}{\let\fonte\cpsource}{}%
1128 % \newcoppefloat{env}{caption name}{list title}: declare an extra caption-on-top
1129 % float (e.g. "picture") that supports \source and gains a \listof<env>s command.
1130 \newcommand{\newcoppefloat}[3]{%
1131   {\floatstyle{plaintop}\newfloat{#1}{htbp}{lo#1}}%
1132   \floatname{#1}{#2}%
1133   \expandafter\newcommand\csname listof#1s\endcsname{%
1134     \chapter*{#3}\addcontentsline{toc}{chapter}{#3}%
1135     \@mkboth{\MakeUppercase{#3}}{\MakeUppercase{#3}}%
1136     \@starttoc{lo#1}}}%
1137 % --- Footnotes (manual2024 / R16, R8, R21): single spacing and a 5 cm rule
1138 %   (filete) from the left margin; footnotesize is the book default. -----
1139 \let\coppe@orig@footnotetext\@footnotetext
1140 \long\def\@footnotetext#1{\coppe@orig@footnotetext{\singlespacing#1}}
1141 \renewcommand\footnoterule{\kern-3\p@\hrule\@width 5cm\kern 2.6\p@}
1142 % --- alineas: list keyed by lowercase letters a) b) c); subitems with "--"
1143 %   (manual2024 2.6 / proposta P24). -----
1144 \RequirePackage{enumitem}
1145 \newlist{alineas}{enumerate}{2}
1146 \setlist{alineas,1}{label=\alph*},leftmargin=1.8em,itemsep=0pt}
1147 \setlist{alineas,2}{label=--,leftmargin=1.8em,itemsep=0pt}
1148 % --- Page numbering: arabic in the top OUTER corner on textual pages
1149 %   (manual2024 2.7 / proposta P25-P26). Pre-textual numbering is set by
1150 %   \frontmatter (unnumbered by default; roman with the numeraisromanos
1151 %   class option). -----
1152 \RequirePackage{fancyhdr}
1153 % \headheight is set to 15pt by geometry up above (see the geometry call),
1154 % which is what fancyhdr needs (>= 14.49998pt) for the running head.
1155 \fancypagestyle{coppe}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
1156 \renewcommand{\headrulewidth}{0pt}}
1157 % Blank pages inserted by \cleardoublepage (two-sided) must be empty: no page
1158 % number may leak onto them (e.g. the verso right after the cover).
1159 \renewcommand{\cleardoublepage}{%
1160   \clearpage
1161   \if@twoside
1162     \ifodd\c@page\else
1163       \hbox{}\thispagestyle{empty}\newpage
1164       \if@twocolumn\hbox{}\newpage\fi
1165     \fi
1166     \fi}
1167 \def\CoppeTeX{{\rm C\kern-.05em{\sc o\kern-.025em p\kern-.025em
1168 p\kern-.025em e}}\kern-.08em
1169 T\kern-.1667em\lower.5ex\hbox{E}\kern-.125emX\spacefactor1000}
1170 \newboolean{maledoc}
1171 \setboolean{maledoc}{false}
1172 %
1173 % Class options.
1174 % Multilingual model (CoppeTeX 4.x, branch nlinguas). The class holds three
1175 % fixed language slots:

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1176 % * main      -- the language of the body, captions, and the main abstract.
1177 %               Set by a class option: brazilian (default), english,
1178 %               spanish, french, italian, or any user-supplied language
1179 %               that ships a coppe-lang-<name>.def file.
1180 % * foreign    -- the language used by the foreignabstract environment and
1181 %               by \foreign@... registers. Defaults to english (or to
1182 %               brazilian when main=english) for back-compat.
1183 % * third      -- optional, loaded on demand with \usecoppelanguage{<name>}
1184 %               in the preamble (for citations or quotations in a third
1185 %               language). Has no class option of its own.
1186 % Note: the legacy \if@english boolean (defined and toggled here in
1187 % v3.x for backward compatibility) was removed in v4.0 -- nothing
1188 % inside the class or in the biblatex style files (.bbx/.cbx/.dbx/.lbr)
1189 % tests it. Code that needs to branch on the main language should test
1190 % \ifx\coppe@mainlang\coppe@englishstr instead.
1191 \def\coppe@mainlang{brazilian}
1192 \def\coppe@foreignlang{english}
1193 \DeclareOption{brazilian}{%
1194   \def\coppe@mainlang{brazilian}\def\coppe@foreignlang{english}}
1195 \DeclareOption{english}{%
1196   \def\coppe@mainlang{english}\def\coppe@foreignlang{brazilian}}
1197 \DeclareOption{spanish}{%
1198   \def\coppe@mainlang{spanish}\def\coppe@foreignlang{english}}
1199 \DeclareOption{french}{%
1200   \def\coppe@mainlang{french}\def\coppe@foreignlang{english}}
1201 \DeclareOption{italian}{%
1202   \def\coppe@mainlang{italian}\def\coppe@foreignlang{english}}
1203 \DeclareOption{msc}{%
1204   \newcommand{\@degree}{M.Sc.}
1205   \newcommand{\@degreename}{Mestrado}
1206   \newcommand{\local@degname}{Mestre}
1207   \newcommand{\foreign@degname}{Master}
1208   \newcommand\local@doctype{Disserta{\c c}{\~ a}o}
1209   \newcommand\foreign@doctype{Dissertation}
1210 }
1211 \DeclareOption{dsccexam}{%
1212   \newcommand{\@degree}{D.Sc.}
1213   \newcommand{\@degreename}{Doutorado}
1214   \newcommand{\local@degname}{Doutor}
1215   \newcommand{\foreign@degname}{Doctor}
1216   \setboolean{maledoc}{true}
1217   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1218   \newcommand\foreign@doctype{Qualifying Exam}
1219 }
1220 \DeclareOption{mscexam}{%
1221   \newcommand{\@degree}{M.Sc.}
1222   \newcommand{\@degreename}{Mestrado}
1223   \newcommand{\local@degname}{Mestre}
1224   \newcommand{\foreign@degname}{Master}
1225   \setboolean{maledoc}{true}
1226   \newcommand\local@doctype{Exame de Qualifica{\c c}{\~ a}o}
1227   \newcommand\foreign@doctype{Qualifying Exam}
1228 }
1229 \DeclareOption{dsc}{%

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1230 \newcommand{\@degree}{D.Sc.}
1231 \newcommand{\@degreename}{Doutorado}
1232 \newcommand{\local@degreename}{Doutor}
1233 \newcommand{\foreign@degreename}{Doctor}
1234 \newcommand\local@doctype{Tese}
1235 \newcommand\foreign@doctype{Thesis}
1236 }
1237 \newif\if@coppenumeric\@coppenumericfalse
1238 \DeclareOption{numbers}{\@coppenumerictrue}
1239 % numeraisromanos: show roman numerals on the pre-textual pages. Default: the
1240 % pre-textual pages are counted but not numbered (manual2024 2.7).
1241 \newif\if@copperoman\@copperomanfalse
1242 \DeclareOption{numeraisromanos}{\@copperomantrue}

Default line spacing is one-and-a-half; the doublespacing option switches to dou-
ble spacing. Since setspace is loaded at the end of the class (after hyperref), the
actual \onehalfspacing/\doublespacing call is deferred: the option only flips a
flag, which the late-load block reads.
1243 \newif\if@coppedoublespace\@coppedoublespacefalse
1244 \DeclareOption{doublespacing}{\@coppedoublespacetrue}

1245 \ProcessOptions\relax
1246 % Built-in language names (used by the loader and the dispatcher below).
1247 \def\coppe@@brazilianstr{brazilian}
1248 \def\coppe@@englishstr{english}
1249 % Load Babel with the foreign language first and the main language last
1250 % (Babel treats the LAST option in the list as the main language). With
1251 % pt-main this produces [english,brazilian]{babel} and with en-main
1252 % [brazilian,english] -- identical to the historical \if@english calls.
1253 \edef\coppe@babelopts{\coppe@foreignlang,\coppe@mainlang}
1254 \expandafter\RequirePackage\expandafter[\coppe@babelopts]{babel}
1255 % --- Multilingual string dispatcher -----
1256 % \copperdefstring{<lang>}{<key>}{<value>} adds one entry to the table for
1257 % language <lang>. The three retrievers below look strings up either at the
1258 % current babel language (\coppestring -- follows \selectlanguage), or pinned
1259 % to the main slot (\coppemainstring) or the foreign slot (\coppeforeignstring).
1260 \providecommand{\copperdefstring}[3]{%
1261   \expandafter\gdef\csname coppe@str@#1@#2\endcsname{#3}}
1262 \providecommand{\coppestring}[1]{\@nameuse{coppe@str@language @#1}}
1263 \providecommand{\coppemainstring}[1]{\@nameuse{coppe@str@\coppe@mainlang @#1}}
1264 \providecommand{\coppeforeignstring}[1]{\@nameuse{coppe@str@\coppe@foreignlang @#1}}
1265 % Load an extra language pack at any point in the preamble (third-language
1266 % slot for citations or quotations beyond the main+foreign pair).
1267 \providecommand{\usecoppelanguage}[1]{%
1268   \def\coppe@@cur{#1}%
1269   \ifx\coppe@@cur\coppe@@brazilianstr\else
1270     \ifx\coppe@@cur\coppe@@englishstr\else
1271       \InputIfFileExists{coppe-lang-#1.def}{}
1272       {\ClassWarning{coppe}{Language pack coppe-lang-#1.def not found}}%
1273     \fi\fi}
1274 % \titlein{<lang>}{<text>}: record the title in any language. Will be
1275 % consulted by the cover/folha renderers once they are migrated in step 2.
1276 % In step 1 it is just a no-op store -- the historical \title / \foreigntitle
1277 % still drive every visible site, so behavior for pt/en is unchanged.
1278 \providecommand{\titlein}[2]{%

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1279 \expandafter\gdef\csname coppe@title@#1\endcsname{#2}}
1280 % --- Built-in string tables: brazilian + english. The values mirror the
1281 % literals currently scattered through the class; step-1 only populates the
1282 % tables -- no site reads them yet, so the rendered output is unchanged. ----
1283 \copperdefstring{brazilian}{appendix}{Ap\~endice}
1284 \copperdefstring{english}{appendix}{Appendix}
1285 \copperdefstring{brazilian}{annex}{Anexo}
1286 \copperdefstring{english}{annex}{Annex}
1287 \copperdefstring{brazilian}{frame}{Quadro}
1288 \copperdefstring{english}{frame}{Frame}
1289 \copperdefstring{brazilian}{listframe}{Lista de Quadros}
1290 \copperdefstring{english}{listframe}{List of Frames}
1291 \copperdefstring{brazilian}{source}{Fonte}
1292 \copperdefstring{english}{source}{Source}
1293 \copperdefstring{brazilian}{program}{Programa}
1294 \copperdefstring{english}{program}{Program}
1295 \copperdefstring{brazilian}{listprogram}{Lista de Programas}
1296 \copperdefstring{english}{listprogram}{List of Programs}
1297 \copperdefstring{brazilian}{references}{Refer\~encias}
1298 \copperdefstring{english}{references}{References}
1299 \copperdefstring{brazilian}{in:}{em}
1300 \copperdefstring{english}{in:}{in}
1301 \copperdefstring{brazilian}{art}{Ilustra\c{c}\~ao}
1302 \copperdefstring{english}{art}{Art}
1303 \copperdefstring{brazilian}{platform}{Plataforma}
1304 \copperdefstring{english}{platform}{Platform}
1305 \copperdefstring{brazilian}{directedby}{Dire\c{c}\~ao}
1306 \copperdefstring{english}{directedby}{Directed by}
1307 \copperdefstring{brazilian}{producedby}{Produ\c{c}\~ao}
1308 \copperdefstring{english}{producedby}{Produced by}
1309 \copperdefstring{brazilian}{listabbreviation}{Lista de Abreviaturas e Siglas}
1310 \copperdefstring{english}{listabbreviation}{List of Abbreviations and Acronyms}
1311 \copperdefstring{brazilian}{listsymbol}{Lista de S{\~i}mbolos}
1312 \copperdefstring{english}{listsymbol}{List of Symbols}
1313 \copperdefstring{brazilian}{glossary}{Gloss{\~a}rio}
1314 \copperdefstring{english}{glossary}{Glossary}
1315 \copperdefstring{brazilian}{advisor}{Orientador}
1316 \copperdefstring{english}{advisor}{Advisor}
1317 \copperdefstring{brazilian}{advisors}{Orientadores}
1318 \copperdefstring{english}{advisors}{Advisors}
1319 \copperdefstring{brazilian}{dept}{Programa}
1320 \copperdefstring{english}{dept}{Department}
1321 \copperdefstring{brazilian}{approvedmale}{Examinado por}
1322 \copperdefstring{brazilian}{approvedfemale}{Examinada por}
1323 \copperdefstring{english}{approvedmale}{Examined by}
1324 \copperdefstring{english}{approvedfemale}{Examined by}
1325 % Institutional names: always Portuguese on the cover/folha (UFRJ template).
1326 \copperdefstring{brazilian}{universityname}{Universidade Federal do Rio de Janeiro}
1327 \copperdefstring{english}{universityname}{Universidade Federal do Rio de Janeiro}
1328 \copperdefstring{brazilian}{cityname}{Rio de Janeiro}
1329 \copperdefstring{english}{cityname}{Rio de Janeiro}
1330 \copperdefstring{brazilian}{statename}{RJ}
1331 \copperdefstring{english}{statename}{RJ}
1332 \copperdefstring{brazilian}{countryname}{Brasil}

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1333 \copperdefstring{english} {countryname} {Brasil}
1334 % Month names.
1335 \copperdefstring{brazilian}{monthname1} {Janeiro}
1336 \copperdefstring{brazilian}{monthname2} {Fevereiro}
1337 \copperdefstring{brazilian}{monthname3} {Mar{\c c}o}
1338 \copperdefstring{brazilian}{monthname4} {Abril}
1339 \copperdefstring{brazilian}{monthname5} {Maio}
1340 \copperdefstring{brazilian}{monthname6} {Junho}
1341 \copperdefstring{brazilian}{monthname7} {Julho}
1342 \copperdefstring{brazilian}{monthname8} {Agosto}
1343 \copperdefstring{brazilian}{monthname9} {Setembro}
1344 \copperdefstring{brazilian}{monthname10}{Outubro}
1345 \copperdefstring{brazilian}{monthname11}{Novembro}
1346 \copperdefstring{brazilian}{monthname12}{Dezembro}
1347 \copperdefstring{english}{monthname1} {January}
1348 \copperdefstring{english}{monthname2} {February}
1349 \copperdefstring{english}{monthname3} {March}
1350 \copperdefstring{english}{monthname4} {April}
1351 \copperdefstring{english}{monthname5} {May}
1352 \copperdefstring{english}{monthname6} {June}
1353 \copperdefstring{english}{monthname7} {July}
1354 \copperdefstring{english}{monthname8} {August}
1355 \copperdefstring{english}{monthname9} {September}
1356 \copperdefstring{english}{monthname10}{October}
1357 \copperdefstring{english}{monthname11}{November}
1358 \copperdefstring{english}{monthname12}{December}
1359 % algorithm2e package option for this language (empty -> english default).
1360 \copperdefstring{brazilian}{algorithm2eopt}{portuguese}
1361 \copperdefstring{english} {algorithm2eopt}{}
1362 % Babel package name for the foreignabstract \begin{otherlanguage}{...}.
1363 \copperdefstring{brazilian}{babelname}{brazilian}
1364 \copperdefstring{english} {babelname}{english}
1365 % Auto-load language packs for any non-built-in language listed in babelopts.
1366 % Built-ins (brazilian, english) ship their string tables inside coppe.cls
1367 % (defined just above) and need no extra file; others (spanish/french/italian
1368 % /...) ship as coppe-lang-<name>.def alongside the class. The dispatcher
1369 % must already be defined when these run, so the loader sits AFTER the tables.
1370 \@for\coppe@@lang:=\coppe@babelopts\do{%
1371   \edef\coppe@@cur{\coppe@@lang}%
1372   \ifx\coppe@@cur\coppe@@brazilianstr\else
1373     \ifx\coppe@@cur\coppe@@englishstr\else
1374       \InputIfFileExists{coppe-lang-\coppe@@cur.def}
1375       {\ClassInfo{coppe}{Loaded language pack coppe-lang-\coppe@@cur.def}}
1376       {\ClassError{coppe}
1377         {Missing language pack 'coppe-lang-\coppe@@cur.def'.}
1378         {Install it next to coppe.cls, or pick a built-in language.}}}%
1379   \fi\fi}
1380 % Quotation marks: csquotes gives the language-aware \enquote; \citacao is its
1381 % Portuguese alias for short, inline quotations (proposta P3).
1382 \RequirePackage{csquotes}
1383 \newcommand{\citacao}[1]{\enquote{#1}}
1384 % --- Code listings & algorithms (TOD03 "tratar listagens"): the listings
1385 % package gives the "Programa" listing (caption on top, ABNT R46) with
1386 % \listofprogramas; algorithm2e gives algorithms. Names are babel-aware and

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1387 % the language styles below are based on the user's listagens.tex. -----
1388 \RequirePackage{xcolor}
1389 \RequirePackage{listings}
1390 \RequirePackage{fancyvrb}
1391 \DeclareFixedFont{\ttb}{T1}{txtt}{bx}{n}{10}
1392 \DeclareFixedFont{\ttm}{T1}{txtt}{m}{n}{10}
1393 \lstset{captionpos=t,extendedchars=true,basicstyle=\ttfamily,inputencoding=utf8,%
1394 literate=%
1395 {ã}{\~{a}}1 {ä}{\~{a}}1 {å}{\~{a}}1 {ä}{\~{a}}1 {é}{\~{e}}1%
1396 {ê}{\~{e}}1 {ë}{\~{e}}1 {î}{\~{i}}1 {ï}{\~{i}}1 {ô}{\~{o}}1%
1397 {ó}{\~{o}}1 {ö}{\~{o}}1 {û}{\~{u}}1 {ü}{\~{u}}1 {ç}{\c{c}}1%
1398 {ç}{\c{C}}1 {À}{\~{A}}1 {Á}{\~{A}}1 {Â}{\~{A}}1 {Ã}{\~{A}}1 {É}{\~{E}}1%
1399 {Ê}{\~{E}}1 {Î}{\~{I}}1 {Ï}{\~{I}}1 {Ô}{\~{O}}1 {Õ}{\~{O}}1%
1400 {Ü}{\~{U}}1%
1401 }
1402 % Babel-aware listing names, set after listings' own babel setup.
1403 \AtBeginDocument{%
1404 \renewcommand\lstlistingname{\coppestring{program}}%
1405 \renewcommand\lstlistlistingname{\coppestring{listprogram}}%
1406 % \listofprogramas in the coppe list style (heading + ToC + running head).
1407 \newcommand\listofprogramas{%
1408 \chapter*{\lstlistlistingname}%
1409 \addcontentsline{toc}{chapter}{\lstlistlistingname}%
1410 \@mkboth{\MakeUppercase\lstlistlistingname}{\MakeUppercase\lstlistlistingname}%
1411 \@starttoc{lol}}
1412 % Per-language styles + environments (python, java, xml, html, prolog).
1413 \newcommand\pythonstyle{\lstset{language=Python,basicstyle=\ttm,numbers=left,%
1414 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1415 morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1416 emph={MyClass,__init__},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1417 stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1418 extendedchars=true,firstnumber=auto,inputencoding=utf8,breaklines=true,%
1419 escapechar=',postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}%
1420 \lstnewenvironment{python}[1][\pythonstyle\lstset{#1}}%
1421 \csname\@lst @SetFirstNumber\endcsname\csname\@lst @SaveFirstNumber\endcsname%
1422 \newcommand\pythoninline[1][\pythonstyle\lstinline!#1!}%
1423 \newcommand\pythonexternal[2][\pythonstyle\lstinputlisting{#1}{#2}}%
1424 \newcommand\xmlstyle{\lstset{language=XML,basicstyle=\ttm,numbers=left,%
1425 numberstyle=\small\ttfamily,numbersep=0.5em,%
1426 morekeywords={xs:schema,xs:element,xs:sequence,xs:complexType},%
1427 keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1428 emph={name,type,attributeFormDefault,elementFormDefault,xmlns:xs},%
1429 emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1430 showstringspaces=false,extendedchars=true,inputencoding=utf8,breaklines=true,%
1431 postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}%
1432 \lstnewenvironment{xml}[1][\xmlstyle\lstset{#1}}%
1433 \newcommand\htmlstyle{\lstset{language=HTML,basicstyle=\ttm,numbers=left,%
1434 numberstyle=\small\ttfamily,numbersep=0.5em,%
1435 keywordstyle=\ttb\color{rgb}{0,0,0.5},emphstyle=\ttb\color{rgb}{0.6,0,0},%
1436 stringstyle=\color{rgb}{0,0.5,0},frame=tb,showstringspaces=false,%
1437 extendedchars=true,inputencoding=utf8,breaklines=true,%
1438 postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}%
1439 \lstnewenvironment{html}[1][\htmlstyle\lstset{#1}}%
1440 \newcommand\prologstyle{\lstset{language=Prolog,basicstyle=\ttm,numbers=left,%

```

```

1441 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1442 keywordstyle=\ttb\color{rgb}{0,0,0.5},emph={MyClass,__init__},%
1443 emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1444 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1445 breaklines=true,escapechar=',%
1446 postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}
1447 \lstnewenvironment{prolog}[1][\prologstyle\lstset{#1}%
1448 \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
1449 \newcommand\javastyle{\lstset{language=Java,basicstyle=\ttm,numbers=left,%
1450 numberstyle=\small\ttfamily,numbersep=0.5em,commentstyle=\color{purple},%
1451 morekeywords={self,as},keywordstyle=\ttb\color{rgb}{0,0,0.5},%
1452 emphstyle=\ttb\color{rgb}{0.6,0,0},stringstyle=\color{rgb}{0,0.5,0},frame=tb,%
1453 showstringspaces=false,extendedchars=true,firstnumber=auto,inputencoding=utf8,%
1454 breaklines=true,escapechar=',%
1455 postbreak=\mbox{\textcolor{red}{$\hookrightarrow$}\space}}
1456 \lstnewenvironment{java}[1][\javastyle\lstset{#1}%
1457 \csname\@lst @SetFirstNumber\endcsname}{\csname\@lst @SaveFirstNumber\endcsname}
1458 \newcommand\javainline[1]{\javastyle\lstinline!#1!}%
1459 \newcommand\javaexternal[2][\javastyle\lstinputlisting{#1}{#2}}
1460 % fancyvrb's Verbatim environment is available for program output; the
1461 % \DefineVerbatimEnvironment{gxoutput}/{gxoutputs} aliases that used to live
1462 % here were author-personal (gx prefix) and were removed in v4.0. Use, for
1463 % example,
1464 % \begin{Verbatim}[frame=lines,numbers=left,numbersep=0.5em]
1465 % ...
1466 % \end{Verbatim}
1467 % (or \DefineVerbatimEnvironment{...}{Verbatim}{...} in the manuscript preamble).
1468 \renewcommand\theFancyVerbLine{\small\ttfamily\arabic{FancyVerbLine}}
1469 % Algorithms: algorithm2e, language driven by the main-slot \coppemainstring
1470 % lookup (key "algorithm2eopt"). Empty means use algorithm2e's default
1471 % language (English) -- needed because algorithm2e errors on an empty option.
1472 \edef\coppe@algotlang{\coppemainstring{algorithm2eopt}}
1473 \ifx\coppe@algotlang\empty
1474 \RequirePackage[linesnumbered,lined,algochapter,ruled]{algorithm2e}
1475 \else
1476 \expandafter\RequirePackage\expandafter[\coppe@algotlang,linesnumbered,lined,algochapter,ruled]{algorithm2e}
1477 \fi
1478 \SetKwFor{Para}{para}{faça}{fim do para}
1479 \SetKwFor{Enquanto}{enquanto}{faça}{fim do enquanto}
1480 % setspace + hyperref + biblatex are loaded at the very end of the class
1481 % (see the block just before %</class>), so their hooks see every macro the
1482 % class defines, and the load order setspace -> hyperref -> biblatex is
1483 % respected.

```

`\department` This macro is used to set the author's affiliation. There are twelve options which correspond to all academic units at COPPE/UFRJ. It defines the current and the foreign names of these units.

```

1484 \newcommand\department[1]{%
1485 \ifthenelse{\equal{#1}{PEB}}{
1486 \global\def\local@deptname{Engenharia Biom{\'e}dica}
1487 \global\def\foreign@deptname{Biomedical Engineering}}{
1488 \ifthenelse{\equal{#1}{PEC}}{
1489 \global\def\local@deptname{Engenharia Civil}
1490 \global\def\foreign@deptname{Civil Engineering}}{

```

```

1491 \ifthenelse{\equal{#1}{PEE}}
1492   {\global\def\local@deptname{Engenharia El{\` e}trica}
1493    \global\def\foreign@deptname{Electrical Engineering}}{}
1494 \ifthenelse{\equal{#1}{PEM}}
1495   {\global\def\local@deptname{Engenharia Mec{\` a}nica}
1496    \global\def\foreign@deptname{Mechanical Engineering}}{}
1497 \ifthenelse{\equal{#1}{PEMM}}
1498   {\global\def\local@deptname{Engenharia Metal{\` u}rgica e de Materiais}
1499 \global\def\foreign@deptname{Metallurgical and Materials Engineering}}{}
1500 \ifthenelse{\equal{#1}{PEN}}
1501   {\global\def\local@deptname{Engenharia Nuclear}
1502    \global\def\foreign@deptname{Nuclear Engineering}}{}
1503 \ifthenelse{\equal{#1}{PEN0}}
1504   {\global\def\local@deptname{Engenharia Oce{\` a}nica}
1505    \global\def\foreign@deptname{Ocean Engineering}}{}
1506 \ifthenelse{\equal{#1}{PPE}}
1507   {\global\def\local@deptname{Planejamento Energ{\` e}tico}
1508    \global\def\foreign@deptname{Energy Planning}}{}
1509 \ifthenelse{\equal{#1}{PEP}}
1510   {\global\def\local@deptname{Engenharia de Produ{\c c}{\` a}o}
1511    \global\def\foreign@deptname{Production Engineering}}{}
1512 \ifthenelse{\equal{#1}{PEQ}}
1513   {\global\def\local@deptname{Engenharia Qu{\` i}mica}
1514    \global\def\foreign@deptname{Chemical Engineering}}{}
1515 \ifthenelse{\equal{#1}{PESC}}
1516   {\global\def\local@deptname{Engenharia de Sistemas e Computa{\c c}{\` a}o}
1517    \global\def\foreign@deptname{Systems Engineering and Computer Science}}{}
1518 \ifthenelse{\equal{#1}{PET}}
1519   {\global\def\local@deptname{Engenharia de Transportes}
1520    \global\def\foreign@deptname{Transportation Engineering}}{}
1521 \ifthenelse{\equal{#1}{PENT}}
1522   {\global\def\local@deptname{Engenharia de Nanotecnologia}
1523    \global\def\foreign@deptname{Nanotechnology Engineering}}{}
1524 }

```

`\title` Used to enter the title in Brazilian Portuguese. Mirrors into `\coppe@title@brazilian` so that the per-language title slot used by the cover/folha renderers (`\coppe@selecttitle`) picks the right text for a pt-main thesis.

```

1525 \renewcommand\title[1]{%
1526   \global\def\local@title{#1}%
1527   \expandafter\gdef\csname coppe@title@brazilian\endcsname{#1}%
1528 }

```

`\foreigntitle` Used to enter the foreign title. Also mirrors into `\coppe@title@english` so that the cover renders the right text for an english-main thesis.

```

1529 \newcommand\foreigntitle[1]{%
1530   \global\def\foreign@title{#1}%
1531   \expandafter\gdef\csname coppe@title@english\endcsname{#1}%
1532 }

```

`\coppe@selecttitle` Expands to the title stored for the current main language. For pt-main this is `\local@title`; for en-main `\foreign@title`; for any other main language the user must call `\titlein{<lang>}{...}` to populate it.

```

1533 \newcommand\coppe@selecttitle{\@nameuse{coppe@title@\coppe@mainlang}}

```

`\advisor` Defines globally the title, name and academic degree of the advisors.

```
1534 \newcount\@advisor\@advisor0
1535 \newcommand\advisor[4]{%
1536   \global\@namedef{CoppeAdvisorTitle:\expandafter\the\@advisor}{#1}
1537   \global\@namedef{CoppeAdvisorName:\expandafter\the\@advisor}{#2}
1538   \global\@namedef{CoppeAdvisorSurname:\expandafter\the\@advisor}{#3}
1539   \global\@namedef{CoppeAdvisorDegree:\expandafter\the\@advisor}{#4}
1540   \global\advance\@advisor by 1
1541   \ifnum\@advisor>1
1542     \renewcommand\local@advisorstring{Orientadores}
1543     \renewcommand\foreign@advisorstring{Advisors}
1544   \fi
1545 }
```

`\examiner`

```
1546 \newcount\@examiner\@examiner0
1547 \newcommand\examiner[3]{%
1548   \global\@namedef{CoppeExaminer:\expandafter\the\@examiner}{#1\ #2}
1549   \global\advance\@examiner by 1
1550 }
```

`\author` It was redefined to allow the identification of the author's first names and surname.

```
1551 \renewcommand\author[2]{%
1552   \global\def\@authname{#1}
1553   \global\def\@authsurn{#2}
1554 }
```

`\date` Sets the *document's* nominal month and year (used by the cover, the folha de rosto, and the catalog). Stores them in private slots (`\coppe@docmonth` / `\coppe@docyear`) rather than mutating TeX's `\month` / `\year` primitives, which would clobber `\today`.

```
1555 \renewcommand\date[2]{%
1556   \def\coppe@docmonth{#1}%
1557   \def\coppe@docyear{#2}}
```

`\local@monthname`

```
1558 \newcommand\local@monthname{\ifcase\coppe@docmonth\or
1559   Janeiro\or Fevereiro\or Mar{\c c}\or Abril\or Maio\or Junho\or
1560   Julho\or Agosto\or Setembro\or Outubro\or Novembro\or Dezembro\fi}
```

`\foreign@monthname`

```
1561 \newcommand\foreign@monthname{\ifcase\coppe@docmonth\or
1562   January\or February\or March\or April\or May\or June\or
1563   July\or August\or September\or October\or November\or December\fi}
```

`\keyword`

```
1564 \newcounter{keywords}
1565 \newcommand\keyword[1]{%
1566   \global\@namedef{CoppeKeyword:\expandafter\the\c@keywords}{#1}
1567   \global\addtocounter{keywords}{1}
1568 }
```

`\freeconfig` This command allows easy changing of core class parameters.

```

1569 \newcommand\freeconfig[6]{%
1570 \providecommand\@degree{}
1571 \renewcommand\@degree{#1}
1572 \providecommand\@degreename{}
1573 \renewcommand\@degreename{#2}
1574 \providecommand\local@degname{}
1575 \renewcommand\local@degname{#3}
1576 \providecommand\foreign@degname{}
1577 \renewcommand\foreign@degname{#4}
1578 \providecommand\local@doctype{}
1579 \renewcommand\local@doctype{#5}
1580 \providecommand\foreign@doctype{}
1581 \renewcommand\foreign@doctype{#6}%
1582 }
1583 % \end{macrocode}
1584 % \end{macro}
1585 %
1586 % \begin{macro}{\frontmatter}
1587 % The number of pages for both frontmatter and mainmatter printed
1588 % in the cataloging details page is computed by means of simple
1589 % \LaTeX\ labels.
1590 % \begin{macrocode}
1591 \renewcommand\frontmatter{%
1592 \cleardoublepage
1593 \@mainmatterfalse
1594 \pagenumbering{roman}
1595 \thispagestyle{empty}
1596 % folha de rosto = counted page 1 (ABNT: the capa is not counted). An odd
1597 % start keeps the page-counter parity aligned with the physical (recto) side,
1598 % so \cleardoublepage lands every pre-textual element on an odd (front) page.
1599 \setcounter{page}{1}
1600 \makefrontpage
1601 \clearpage
1602 \if@copperoman
1603 \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
1604 \renewcommand{\headrulewidth}{0pt}}%
1605 \pagestyle{coppe}%
1606 \else
1607 \fancypagestyle{plain}{\fancyhf{}\renewcommand{\headrulewidth}{0pt}}%
1608 \pagestyle{empty}%
1609 \fi
1610 }
```

`\mainmatter`

```

1611 \renewcommand\mainmatter{%
1612 \coppe@mainBegin
1613 \cleardoublepage
1614 \@mainmattertrue
1615 \fancypagestyle{plain}{\fancyhf{}\fancyhead[R0,LE]{\thepage}}%
1616 \renewcommand{\headrulewidth}{0pt}}%
1617 \pagestyle{coppe}%
1618 \pagenumbering{arabic}}
```

`\backmatter`

```
1619 \renewcommand\backmatter{%
1620   \if@openright
1621     \cleardoublepage
1622   \else
1623     \clearpage
1624   \fi}
1625 %
```

`\coppetexfinalpage` Optional colophon-style closing page intended to face the rear cover. The macro forces a verso (even) page so that, in a two-sided print, the reader sees the colophon on the left and the inside of the rear cover on the right; pushes the text to the bottom of the page with `\vfill`; and types it in Portuguese unconditionally (the colophon is institutional metadata, not body content, so it is rendered in pt-BR regardless of the thesis main language). Three slot macros are user-overridable: `\coppefinalengine` (engine name; auto-detected via `iftex`), `\coppefinalfont` (text body font; defaults to Computer Modern, the L^AT_EX default), and `\coppefinalmanual` (the reference manual title; defaults to the current COPPE/UFRJ norm). Override them before `\coppetexfinalpage` with `\renewcommand` if the document uses a non-default font or follows an earlier edition of the norm.

```
1626 \providecommand{\coppefinalengine}{%
1627   \ifPDFTeX\hologo{pdfLaTeX}\fi
1628   \ifLuaTeX\hologo{LuaLaTeX}\fi
1629   \ifXeTeX\hologo{XeLaTeX}\fi}
1630 \providecommand{\coppefinalfont}{Computer Modern (padr\~ao \LaTeX{})}
1631 \providecommand{\coppefinalmanual}{%
1632   \emph{Norma para a Elabora\c c\~ao Gr\'afica de Teses e Disserta\c c\~oes
1633     da COPPE/UFRJ}}
1634 \newcommand{\coppetexfinalpage}{%
1635   \clearpage
1636   % Force the colophon onto a verso (even) page so it faces the rear cover.
1637   % If the current page is odd (recto), fill it with an empty verso first.
1638   \if@twoside
1639     \ifodd\c@page
1640       \hbox{}\thispagestyle{empty}\newpage
1641     \fi
1642   \fi
1643   \thispagestyle{empty}%
1644   \null\vfill
1645   \begin{otherlanguage}{brazilian}%
1646     \begin{center}
1647       {\footnotesize\setstretch{1.0}}%
1648       \textsc{Colof\~ao}\par
1649       \vspace{0.7em}
1650       \begin{minipage}{0.75\textwidth}\centering
1651         Este documento foi composto em \LaTeX{} com \coppefinalengine{}
1652         em \today, na fonte \coppefinalfont. A bibliografia foi processada
1653         por Bib\LaTeX{} e Biber. Foi utilizada a classe \CoppeTeX,
1654         vers\~ao\~ao\coppe@version, dispon\'i vel no reposit\'orio GitHub
1655         \mbox{\url{https://github.com/COPPE-UFRJ/CoppeTeX}}, em
1656         conformidade com a \coppefinalmanual.
1657       \end{minipage}\par

```

```

1658 \end{center}
1659 \end{otherlanguage}%
1660 \vspace*{2cm}%
1661 \clearpage}

```

`\makecover` The ABNT cover (capa, NBR 14724) suggested in the manual's Anexo A. It is the institution block — the university, the full COPPE name, and the graduate program taken from `\department` — followed by the author, the title, and the city and year, all centred and uppercased. The optional UFRJ and COPPE logos head the page. The capa is not counted: it precedes the (roman) front matter and is produced before the `titlepage` (folha de rosto).

```

1662 \newcommand\makecover{%
1663 \begin{titlepage}%
1664 \centering
1665 \makebox[\linewidth]{%
1666 \includegraphics[height=2cm]{ufrj-logo}\hfill
1667 \includegraphics[height=2cm]{coppe-logo}}\par
1668 \vspace{1.5cm}
1669 {\singlespacing
1670 \MakeUppercase{\local@universityname}\par
1671 \nohyphens{\MakeUppercase{Instituto Alberto Luiz Coimbra de
1672 P{\~ o}s-Gradua{\c c}{\~ a}o e Pesquisa de Engenharia (COPPE)}}\par
1673 \MakeUppercase{Programa de \local@deptname}\par}
1674 \vspace{\stretch{2}}
1675 \textbf{\MakeUppercase{\@authname\ \@authsur}}\par
1676 \vspace{\stretch{2}}
1677 \nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1678 \vspace{\stretch{3}}
1679 \MakeUppercase{\local@cityname}\par
1680 \coppe@docyear
1681 \end{titlepage}%
1682 }

```

`\maketitle`

```

1683 \renewcommand\maketitle{%
1684 \pagenumbering{alph}
1685 \ltx@ifpackageloaded{hyperref}{\coppe@hypersetup}{}%
1686 \makecover
1687 \begin{titlepage}
1688 \begin{flushleft}
1689 \vspace*{1.5mm}
1690 \setlength\baselineskip{0pt}
1691 \setlength\parskip{1mm}
1692 \makebox[\linewidth]{%
1693 \includegraphics[height=2cm]{ufrj-logo}\hfill
1694 \includegraphics[height=2cm]{coppe-logo}}
1695 \end{flushleft}
1696 \vspace{1.05cm}
1697 \begin{center}
1698 \nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1699 \vspace*{3cm}
1700 \nohyphens{\@authname\ \@authsur}\par
1701 \end{center}

```

```

1702 \vspace*{2.1cm}
1703 \begin{flushright}
1704 \begin{minipage}{8.45cm}
1705 \frontcover@maintext
1706 \end{minipage}\par
1707 \vspace*{7.5mm}
1708 \nohyphens{%
1709 \begin{tabularx}{8.45cm}[b]{@{}l@{ }}>{\raggedright\arraybackslash}X@{}}
1710 \local@advisorstring: &
1711 \count1=0
1712 \toks@={}
1713 \@whilenum \count1<\@advisor \do{%
1714 \ifcase\count1 % same as \ifnum0=\count1
1715 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1716 \expandafter\endcsname\expandafter\space%
1717 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1718 \else
1719 \toks@=\expandafter\expandafter\expandafter{%
1720 \expandafter\the\expandafter\toks@%
1721 \expandafter&\expandafter\space%
1722 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1723 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1724 }%
1725 \fi
1726 \advance\count1 by 1}
1727 \the\toks@
1728 \end{tabularx}}\par
1729 \end{flushright}
1730 \vspace*{\fill}
1731 \begin{center}
1732 \local@cityname\par
1733 \local@monthname\ de \coppe@docyear
1734 \end{center}
1735 \end{titlepage}
1736 % Ficha catalográfica on the verso of the folha de rosto (NBR 14724, R57):
1737 % the title page is a recto, so this lands on the immediately following verso,
1738 % ahead of the approval sheet produced by \frontmatter.
1739 \ifthenelse{\boolean{maledoc}}{\ifthispagestyle{empty}\makecatalog}%
1740 \global\let\maketitle\relax%
1741 \global\let\and\relax}

1742 \newcommand\makefrontpage{%
1743 \begin{center}
1744 \sloppy\nohyphens{\MakeUppercase{\coppe@selecttitle}}\par
1745 \vspace*{7mm}
1746 {\@authname\ \@authsur}\par
1747 \end{center}\par
1748 \vspace*{4mm}
1749 \frontpage@maintext
1750 \vspace*{16mm}
1751 \nohyphens{%
1752 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }}>{\raggedright\arraybackslash}X@{}}
1753 \local@advisorstring: &
1754 \count1=0

```

```

1755 \toks@={}
1756 \@whilenum \count1<\@advisor \do{%
1757 \ifcase\count1 % same as \ifnum0=\count1
1758 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1759 \expandafter\endcsname\expandafter\space%
1760 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1761 \else
1762 \toks@=\expandafter\expandafter\expandafter{%
1763 \expandafter\the\expandafter\toks@%
1764 \expandafter&\expandafter\space%
1765 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1766 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1767 }%
1768 \fi
1769 \advance\count1 by 1}
1770 \the\toks@
1771 \end{tabularx}\par
1772 \vspace*{20mm}
1773 \noindent\begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1774 Aprovada por: &
1775 \count1=0
1776 \toks@={}
1777 \@whilenum \count1<\@examiner \do{%
1778 \ifcase\count1 % same as \ifnum0=\count1
1779 \toks@=\expandafter{\csname CoppeExaminer:\the\count1%
1780 \expandafter\endcsname\expandafter\\}
1781 \else
1782 \toks@=\expandafter\expandafter\expandafter{%
1783 \expandafter\the\expandafter\toks@%
1784 \expandafter&\expandafter\space%
1785 \csname CoppeExaminer:\the\count1\expandafter\endcsname%
1786 \expandafter\space\\
1787 }%
1788 \fi
1789 \advance\count1 by 1}
1790 \the\toks@
1791 \end{tabularx}\par
1792 \vspace*{\fill}
1793 \frontpage@bottomtext}

1794 \newcommand\coppe@hypersetup{%
1795 \begingroup
1796 % changes to \toks@ and \count@ are kept local;
1797 % it's not necessary for them, but it is usually the case
1798 % for \count1, because the first ten counters are written
1799 % to the DVI file, thus you got lucky because of PDF output
1800 \toks@={} % in this special case not necessary
1801 \count@=0 %
1802 \@whilenum\count@<\value{keywords}\do{%
1803 % * a keyword separator is not necessary,
1804 % if there is just one keyword
1805 % * \csname CoppeKeyword:\the\count@\endcsname must be expanded
1806 % at least once, to get rid of the loop depended \count@
1807 \ifcase\count@ % same as \ifnum0=\count@

```

```

1808     \toks@=\expandafter{\csname CoppeKeyword:\the\count@\endcsname}%
1809     \else
1810     \toks@=\expandafter\expandafter\expandafter{%
1811     \expandafter\the\expandafter\toks@
1812     \expandafter;\expandafter\space
1813     \csname CoppeKeyword:\the\count@\endcsname
1814     }%
1815     \fi
1816     \advance\count@ by 1 %
1817 }%
1818 \edef\x{\endgroup
1819 \noexpand\hypersetup{%
1820 pdfkeywords={\the\toks@}%
1821 }%
1822 }%
1823 \x
1824 \hypersetup{%
1825 pdfauthor={\@authname\ \@authsur},
1826 pdftitle={\local@title},
1827 pdfsubject={\local@doctype\ de \@degreename\ em \local@deptname\ da COPPE/UFRJ},
1828 pdfcreator={LaTeX with CoppeTeX toolkit},
1829 breaklinks={true},
1830 raiselinks={true},
1831 pageanchor={true},
1832 }}

```

`\makecatalog` When the document has illustrations, it is required to insert “: il.” between the number of pages of the textual part and the page dimension. We have created a label to flag the existence of lists of figures. It is checked to be undefined using the plain T_EX command `\@isundefined` (cf. the T_EX FAQ).

```

1833 \newcommand\makecatalog{%
1834   \vspace*{\fill}
1835   \begin{center}
1836     \setlength{\fboxsep}{5mm}
1837     \framebox[120mm][c]{\makebox[5mm][c]{}%
1838     \begin{minipage}[c]{105mm}
1839       \setlength{\parindent}{5mm}
1840       \noindent\sloppy\nohyphens\@authsur,
1841       \nohyphens\@authname\par
1842       \nohyphens{%
1843         \coppe@selecttitle/\@authname\ \@authsur. -- \local@cityname:
1844         UFRJ/COPPE, \coppe@docyear.}\par
1845       \pageref{front:pageno},
1846       \pageref{LastPage}
1847       p.\@ifundefined{r@cat:lofflag}{\pageref{cat:lofflag}}{$29,7$cm.\par
1848       % There is an issue here. When the last entry must be split between lines,
1849       % the spacing between it and the next paragraph becomes smaller.
1850       % Should we manually introduce a fixed space? But how could we know that
1851       % a name was split? Is this happening yet?
1852       \nohyphens{%
1853         \begin{tabularx}{100mm}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1854           \local@advisorstring: &
1855           \count1=0
1856           \toks@={}

```

```

1857 \@whilenum \count1<\@advisor \do{%
1858 \ifcase\count1 % same as \ifnum0=\count1
1859 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1860 \expandafter\endcsname\expandafter\space%
1861 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1862 \else
1863 \toks@=\expandafter\expandafter\expandafter{%
1864 \expandafter\the\expandafter\toks@
1865 \expandafter&\expandafter\space
1866 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1867 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1868 }%
1869 \fi
1870 \advance\count1 by 1}
1871 \the\toks@
1872 \end{tabularx}}\par
1873 \nohyphens{\local@doctype\ ({\MakeLowercase\@degreename}) --
1874 UFRJ/COPPE/Programa de \local@deptname, \coppe@docyear.}\par
1875 Refer{\^ e}ncias Bibliogr{\' a}ficas: p. \pageref{bib:begin} -- \pageref{bib:end}.\par
1876 \count1=0
1877 \count2=1
1878 \nohyphens{\@whilenum \count1<\value{keywords} \do {%
1879 \number\count2. \csname CoppeKeyword:\the\count1 \endcsname.
1880 \advance\count1 by 1
1881 \advance\count2 by 1}
1882 I. \csname CoppeAdvisorSurname:0\endcsname,%
1883 \ \csname CoppeAdvisorName:0\endcsname%
1884 \ifthenelse{\@advisor>1}{\ \emph{et~al.}}{}}.
1885 II. \local@universityname, COPPE, Programa de \local@deptname.
1886 III. T{\' i}tulo.}
1887 \end{minipage}}
1888 \end{center}
1889 \vspace*{\fill}}

```

\dedication

```

1890 \newcommand\dedication[1]{
1891 \gdef\@dedic{#1}
1892 \cleardoublepage
1893 \vspace*{\fill}
1894 \begin{flushright}
1895 \begin{minipage}{60mm}
1896 \raggedleft \it \normalsize \@dedic
1897 \end{minipage}
1898 \end{flushright}}

```

abstract (*env.*) This is a specialization of the abstract in the article standard class.

```

1899 \newenvironment{abstract}{%
1900 \cleardoublepage
1901 \thispagestyle{plain}
1902 \abstract@toptext\par
1903 \vspace*{8.6mm}
1904 \begin{center}
1905 \sloppy\nohyphens{\MakeUppercase\local@title}\par
1906 \vspace*{13.2mm}

```

```

1907 \authname\ \@authsurn \par
1908 \vspace*{7mm}
1909 \local@monthname/\coppe@docyear
1910 \end{center}\par
1911 \vspace*{1.5cm}
1912 \noindent%
1913 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1914 \local@advisorstring: &
1915 \count1=0
1916 \toks@={}
1917 \@whilenum \count1<\@advisor \do{%
1918 \ifcase\count1 % same as \ifnum0=\count1
1919 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1920 \expandafter\endcsname\expandafter\space%
1921 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1922 \else
1923 \toks@=\expandafter\expandafter\expandafter{%
1924 \expandafter\the\expandafter\toks@
1925 \expandafter&\expandafter\space
1926 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1927 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1928 }%
1929 \fi
1930 \advance\count1 by 1}
1931 \the\toks@
1932 \end{tabularx}\par
1933 \vspace*{2mm}
1934 \noindent\local@deptstring: \local@deptname\par
1935 \vspace*{7mm}}{\vspace*{\fill}}

```

foreignabstract (*env.*)

```

1936 \newenvironment{foreignabstract}{%
1937 \cleardoublepage
1938 \thispagestyle{plain}
1939 \begin{otherlanguage}{\coppe@foreignlang}
1940 \foreignabstract@toptext\par
1941 \vspace*{8.6mm}
1942 \begin{center}
1943 \sloppy\nohyphens{\MakeUppercase\foreign@title}\par
1944 \vspace*{13.2mm}
1945 \@authname\ \@authsurn \par
1946 \vspace*{7mm}
1947 \foreign@monthname/\coppe@docyear
1948 \end{center}\par
1949 \vspace*{1.5cm}
1950 \noindent%
1951 \begin{tabularx}{\textwidth}[b]{@{}l@{ }>{\raggedright\arraybackslash}X@{}}
1952 \foreign@advisorstring: &
1953 \count1=0
1954 \toks@={}
1955 \@whilenum \count1<\@advisor \do{%
1956 \ifcase\count1 % same as \ifnum0=\count1
1957 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
1958 \expandafter\endcsname\expandafter\space%

```

```

1959     \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
1960   \else
1961     \toks@=\expandafter\expandafter\expandafter{%
1962       \expandafter\the\expandafter\toks@
1963       \expandafter&\expandafter\space
1964       \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
1965       \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
1966     }%
1967   \fi
1968   \advance\count1 by 1}
1969   \the\toks@
1970 \end{tabularx}\par
1971 \vspace*{2mm}
1972 \noindent\foreign@deptstring: \foreign@deptname\par
1973 \vspace*{7mm}}{%
1974 \end{otherlanguage}
1975 \vspace*{\fill}
1976 \global\let\@author\@empty
1977 \global\let\@date\@empty
1978 \global\let\foreign@title\@empty
1979 \global\let\foreign@title\relax
1980 \global\let\local@title\@empty
1981 \global\let\local@title\relax
1982 \global\let\author\relax
1983 \global\let\author\relax
1984 \global\let\date\relax}

```

brazilianabstract (*env.*) Optional third abstract for non-pt main languages: a Brazilian Portuguese abstract printed in addition to **abstract** (main language) and **foreignabstract** (English by default). Wraps its body in `\begin{otherlanguage}{brazilian}` so the babel typography (hyphenation, quotation marks) is correct regardless of the current main language. Uses the same `\local@...` Portuguese strings the **abstract** env uses today. Meaningful only when main is neither **brazilian** nor **english**; harmless otherwise (the document just won't include it).

```

1985 \newenvironment{brazilianabstract}{%
1986   \cleardoublepage
1987   \thispagestyle{plain}
1988   \begin{otherlanguage}{brazilian}
1989   \abstract@toptext\par
1990   \vspace*{8.6mm}
1991   \begin{center}
1992     \sloppy\nohyphens{\MakeUppercase\local@title}\par
1993     \vspace*{13.2mm}
1994     \@authname\ \@authsurn \par
1995     \vspace*{7mm}
1996     \local@monthname/\coppe@docyear
1997   \end{center}\par
1998   \vspace*{1.5cm}
1999   \noindent%
2000   \begin{tabularx}{\textwidth}[b]{@{}l@{ }>\raggedright\arraybackslash}X@{}}
2001     \local@advisorstring: &
2002     \count1=0
2003     \toks@={}

```

```

2004 \@whilenum \count1<\@advisor \do{%
2005 \ifcase\count1
2006 \toks@=\expandafter{\csname CoppeAdvisorName:\the\count1%
2007 \expandafter\endcsname\expandafter\space%
2008 \csname CoppeAdvisorSurname:\the\count1\endcsname\\}
2009 \else
2010 \toks@=\expandafter\expandafter\expandafter{%
2011 \expandafter\the\expandafter\toks@
2012 \expandafter&\expandafter\space
2013 \csname CoppeAdvisorName:\the\count1\expandafter\endcsname%
2014 \expandafter\space\csname CoppeAdvisorSurname:\the\count1\endcsname\\
2015 }%
2016 \fi
2017 \advance\count1 by 1}
2018 \the\toks@
2019 \end{tabularx}\par
2020 \vspace*{2mm}
2021 \noindent\local@deptstring: \local@deptname\par
2022 \vspace*{7mm}}{%
2023 \end{otherlanguage}
2024 \vspace*{\fill}}

```

\listoffigures

```

2025 \renewcommand\listoffigures{%
2026 \coppe@hasLof
2027 \if@twocolumn
2028 \@restonecoltrue\onecolumn
2029 \else
2030 \@restonecolfalse
2031 \fi
2032 \chapter*{\listfigurename}%
2033 \addcontentsline{toc}{chapter}{\listfigurename}%
2034 \@mkboth{\MakeUppercase\listfigurename}%
2035 {\MakeUppercase\listfigurename}%
2036 \@starttoc{lof}%
2037 \if@restonecol\twocolumn\fi
2038 }

```

\listoftables

```

2039 \renewcommand\listoftables{%
2040 \if@twocolumn
2041 \@restonecoltrue\onecolumn
2042 \else
2043 \@restonecolfalse
2044 \fi
2045 \chapter*{\listtablename}%
2046 \addcontentsline{toc}{chapter}{\listtablename}%
2047 \@mkboth{%
2048 \MakeUppercase\listtablename}%
2049 {\MakeUppercase\listtablename}%
2050 \@starttoc{lot}%
2051 \if@restonecol\twocolumn\fi
2052 }

```

\printlosymbols

```
2053 \newcommand\printlosymbols{%
2054 \renewcommand\glossaryname{\listsymbolname}%
2055 \@input@{\jobname.los}}
```

\makelosymbols

```
2056 \def\makelosymbols{%
2057   \newwrite\@losfile
2058   \immediate\openout\@losfile=\jobname.syx
2059   \newcommand\syml[3][\@bsphack\begingroup
2060   \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
2061       \@sanitize%
2062       \@wrls{\@tempsymb1}{##2}{##3}\typeout%
2063   {Writing index of symbols file \jobname.syx}%
2064   \let\makelosymbols\@empty%
2065 }%
2066 \@onlypreamble\makelosymbols

2067 \AtBeginDocument{%
2068 \ifpackageloaded{hyperref}{%
2069   \newcommand\@wrls[3]{%
2070     \protected@write\@losfile{%
2071       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%
2072     }%
2073     \@esphack}%
2074   \newcommand\@wrls[3]{%
2075     \protected@write\@losfile{%
2076       {\string\indexentry{#1[#2] #3}{\thepage}}%
2077     }%
2078     \@esphack}}}%

```

\printloabbreviations

```
2079 \newcommand\printloabbreviations{%
2080 \renewcommand\glossaryname{\listabbreviationname}%
2081 \@input@{\jobname.lab}}
```

\makeloabbreviations

```
2082 \def\makeloabbreviations{%
2083   \newwrite\@labfile
2084   \immediate\openout\@labfile=\jobname.abx
2085   \newcommand\abbrev[3][\@bsphack\begingroup
2086   \ifstrempy{##1}{\def\@tempsymb1{##2=}}{\def\@tempsymb1{##1=}}%
2087       \@sanitize%
2088       \@wrlab{\@tempsymb1}{##2}{##3}\typeout
2089   {Writing index of abbreviations file \jobname.abx}%
2090   \let\makeloabbreviations\@empty
2091 }%
2092 \@onlypreamble\makeloabbreviations

2093 \AtBeginDocument{%
2094 \ifpackageloaded{hyperref}{%
2095   \newcommand\@wrlab[3]{%
2096     \protected@write\@labfile{%
2097       {\string\indexentry{#1[#2] #3|hyperpage}{\thepage}}%

```

```

2098 \endgroup%
2099 \@esphack}}{%
2100 \newcommand\@wrlab[3]{%
2101 \protected@write\@labfile{%
2102 {\string\indexentry{#1[#2] #3}{\thepage}}}%
2103 \endgroup%
2104 \@esphack}}}%

\newsigla Acronyms (siglas), manual2024 2.8 / R43. \newsigla{key}{SIGLA}{full form}
\sigla declares one; \sigla{key} prints “full form (SIGLA)” on first use and registers it in
the list of abbreviations and acronyms (through \abbrev, so \makeloabbreviations
must be active), then just “SIGLA” on later uses. \sigla*{key} forces the full
form. The first-use flag is set globally so it survives groups (floats, footnotes).

2105 \newcommand\newsigla[3]{%
2106 \@namedef{coppe@sigla@s@#1}{#2}%
2107 \@namedef{coppe@sigla@l@#1}{#3}%
2108 \newcommand\coppe@sigla@reg[1]{%
2109 \@ifundefined{abbrev}}{%
2110 {\abbrev{\@nameuse{coppe@sigla@s@#1}}{\@nameuse{coppe@sigla@l@#1}}}%
2111 \newcommand\coppe@sigla@full[1]{%
2112 \@nameuse{coppe@sigla@l@#1}\space{\@nameuse{coppe@sigla@s@#1}}%
2113 \newcommand\coppe@sigla@undef[1]{%
2114 \PackageWarning{coppe}{Sigla '#1' nao declarada com \string\newsigla}%
2115 \textbf{??#1??}}%
2116 \newcommand\coppe@sigla@first[1]{%
2117 \global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@reg{#1}\coppe@sigla@full{#1}}%
2118 \newcommand\coppe@sigla@norm[1]{%
2119 \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
2120 {\@ifundefined{coppe@sigla@done@#1}{\coppe@sigla@first{#1}}%
2121 {\@nameuse{coppe@sigla@s@#1}}}%
2122 \newcommand\coppe@sigla@star[1]{%
2123 \@ifundefined{coppe@sigla@s@#1}{\coppe@sigla@undef{#1}}%
2124 {\global\@namedef{coppe@sigla@done@#1}{\coppe@sigla@full{#1}}}%
2125 \newcommand\sigla{\@ifstar\coppe@sigla@star\coppe@sigla@norm}

2126 %%% \AtBeginDocument{%
2127 %%% \ifpackageloaded{hyperref}{%
2128 %%% \def\@wrlab#1#2{%
2129 %%% \protected@write\@labfile{%
2130 %%% {\string\indexentry{#1] #2|hyperpage}{\thepage}}}%
2131 %%% \endgroup
2132 %%% \@esphack}}{%
2133 %%% \def\@wrlab#1#2{%
2134 %%% \protected@write\@labfile{%
2135 %%% {\string\indexentry{#1] #2}{\arabic{page}}}%
2136 %%% \endgroup
2137 %%% \@esphack}}

2138 % Catalog-label writers. Two language-related traps must be avoided here:
2139 % 1. The label NAMES contain ':' and '.', which babel makes active in
2140 % some languages (e.g. Brazilian's "shorthand" set). If the write
2141 % argument is tokenised while one of those is active, pdfTeX aborts
2142 % with "<inserted text> \endwrite" because the shorthand expansion
2143 % injects unbalanced braces. \detokenize emits its argument as
2144 % catcode-12 characters regardless of the current activation, so the

```

```

2145 %      label NAMES survive whatever language is current at writer time.
2146 % 2. Spanish babel redefines \@roman (and therefore \roman{...}) to
2147 %      produce small-caps roman numerals via \es@scroman, whose expansion
2148 %      contains an \uppercase group that is unbalanced inside \write. The
2149 %      same crash signature appears. The fix is to bypass LaTeX's
2150 %      \roman/\arabic counter wrappers entirely and use the raw e-TeX
2151 %      primitives \romannumeral and \number directly on the \c@page count
2152 %      register --- both are fully expandable, language-agnostic, and
2153 %      always produce plain ASCII digits or lowercase roman letters.
2154 %      \Roman{page} is left in place because babel-spanish only redefines
2155 %      \@Roman under the (non-default) ucroman option.
2156 \AtBeginDocument{%
2157   \ltx@ifpackageloaded{hyperref}{%
2158     \def\coppe@bibEnd{%
2159       \immediate\write\@auxout{%
2160         \string\newlabel{\detokenize{bib:end}}%
2161         {{{\number\c@page}}{\detokenize{page.}\number\c@page}}}}}%
2162   \def\coppe@bibBegin{%
2163     \immediate\write\@auxout{%
2164       \string\newlabel{\detokenize{bib:begin}}%
2165       {{{\number\c@page}}{\detokenize{page.}\number\c@page}}}}}%
2166   \def\coppe@mainBegin{%
2167     \immediate\write\@auxout{%
2168       \string\newlabel{\detokenize{front:pageno}}%
2169       {{{\Roman{page}}{\detokenize{page.}\romannumeral\c@page}}}}}%
2170   \def\coppe@hasLof{%
2171     \immediate\write\@auxout{%
2172       \string\newlabel{\detokenize{cat:lofflag}}%
2173       {{{\detokenize{:~il.}}{\detokenize{page.}\romannumeral\c@page}}}}}%
2174 }{%
2175   \def\coppe@bibEnd{%
2176     \immediate\write\@auxout{%
2177       \string\newlabel{\detokenize{bib:end}}{{{ \number\c@page}}}}}%
2178   \def\coppe@bibBegin{%
2179     \immediate\write\@auxout{%
2180       \string\newlabel{\detokenize{bib:begin}}{{{ \number\c@page}}}}}%
2181   \def\coppe@mainBegin{%
2182     \immediate\write\@auxout{%
2183       \string\newlabel{\detokenize{front:pageno}}{{{ \Roman{page}}}}}%
2184   \def\coppe@hasLof{%
2185     \immediate\write\@auxout{%
2186       \string\newlabel{\detokenize{cat:lofflag}}{{{ \detokenize{:~il.}}}}}%
2187 }%
2188 }
2189 % The reference list is now produced by biblatex's \printbibliography (see the
2190 % bibliography setup after babel, above). The former thebibliography
2191 % redefinition and its \bibindent are no longer used and have been removed; the
2192 % bib:begin/bib:end catalog labels are now written by \AtBeginBibliography /
2193 % \AtEndBibliography via \coppe@bibBegin and \coppe@bibEnd, defined just above.

```

longquote (*env.*)

```

2194 \newlength{\recuolongquote}%
2195 \setlength{\recuolongquote}{4cm}%
2196 \newenvironment*{longquote}[1][default]{%

```

```

2197 \list{}%
2198 \footnotesize%
2199 \addtolength{\leftskip}{\recuolongquote}%
2200 \item[]%
2201 \singlespacing%
2202 \ifthenelse{\not\equal{#1}{default}}{\itshape\selectlanguage{#1}}{}%
2203 }\endlist}%

2204 \newenvironment{theglossary}{%
2205 \if@twocolumn%
2206 \@restonecoltrue\onecolumn%
2207 \else%
2208 \@restonecolfalse%
2209 \fi%
2210 \@mkboth{\MakeUppercase\glossaryname}%
2211 {\MakeUppercase\glossaryname}%
2212 \chapter*{\glossaryname}%
2213 \addcontentsline{toc}{chapter}{\glossaryname}
2214 \list{}
2215 {\setlength{\listparindent}{0in}%
2216 \setlength{\labelwidth}{1.0in}%
2217 \setlength{\leftmargin}{1.5in}%
2218 \setlength{\labelsep}{0.5in}%
2219 \setlength{\itemindent}{0in}}%
2220 \sloppy}%
2221 {\if@restonecol\twocolumn\fi%
2222 \endlist}
2223 %
2224 \renewenvironment{theindex}{%
2225 \if@twocolumn
2226 \@restonecolfalse
2227 \else
2228 \@restonecoltrue
2229 \fi
2230 \twocolumn[\@makeschapterhead{\indexname}]%
2231 \@mkboth{\MakeUppercase\indexname}%
2232 {\MakeUppercase\indexname}%
2233 \thispagestyle{plain}\parindent\z@
2234 \addcontentsline{toc}{chapter}{\indexname}
2235 \parskip\z@ \@plus .3\p@\relax
2236 \columnseprule \z@
2237 \columnsep 35\p@
2238 \let\item\@idxitem}
2239 {\if@restonecol\onecolumn\else\clearpage\fi}
2240 \newcommand\listabbreviationname{\coppemainstring{listabbreviation}}
2241 \newcommand\listsymbolname{\coppemainstring{listsymbol}}
2242 \newcommand\glossaryname{\coppemainstring{glossary}}
2243 %
2244 \newcommand\local@advisorstring{Orientador}
2245 \newcommand\foreign@advisorstring{Advisor}
2246 \ifthenelse{\boolean{maledoc}}{}%
2247 \newcommand\local@approvedname{Examinado por}%
2248 }\%
2249 \newcommand\local@approvedname{Examinada por}%

```

```

2250 }
2251 \newcommand\local@universityname{Universidade Federal do Rio de Janeiro}
2252 \newcommand\local@deptstring{Programa}
2253 \newcommand\foreign@deptstring{Department}
2254 \newcommand\local@cityname{Rio de Janeiro}
2255 \newcommand\local@statename{RJ}
2256 \newcommand\local@countryname{Brasil}
2257 %
2258 \newcommand\frontcover@maintext{
2259   \sloppy\nohyphens{\local@doctype\ de \@degree\name\
2260   \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2261   ao Programa de P{\ ' o}s-gradua{\c c}{\~ a}o em \local@deptname,
2262   COPPE, da \local@universityname, como parte dos requisitos
2263   necess{\ ' a}rios {\ ' a} obten{\c c}{\~ a}o do t{\ ' i}tulo de
2264   \local@degname\ em \local@deptname.}
2265 }
2266 %
2267 \newcommand\frontpage@maintext{
2268   \noindent {\MakeUppercase\local@doctype}
2269   \ifthenelse{\boolean{maledoc}}{SUBMETIDO}{SUBMETIDA}
2270   \sloppy\nohyphens{AO CORPO DOCENTE DO INSTITUTO ALBERTO LUIZ COIMBRA
2271   DE P{\ ' O}S-GRADUA{\c C}{\~ A}O E PESQUISA DE ENGENHARIA DA
2272   UNIVERSIDADE FEDERAL DO RIO DE JANEIRO COMO PARTE DOS REQUISITOS
2273   NECESS{\ ' A}RIOS PARA A OBTEN{\c C}{\~ A}O DO GRAU DE
2274   {\MakeUppercase\local@degname} EM CI{\^E}NCIAS EM
2275   {\MakeUppercase\local@deptname.\par}}%
2276 }
2277 %
2278 \newcommand\frontpage@bottomtext{%
2279   \begin{center}
2280     {\MakeUppercase{\local@cityname, \local@statename\ -- \local@countryname}}\par
2281     {\MakeUppercase\local@monthname\ DE \coppe@docyear}
2282   \end{center}%
2283 }
2284 %
2285 \newcommand\abstract@toptext{%
2286   \noindent Resumo \ifthenelse{\boolean{maledoc}}{do}{da}
2287   \local@doctype\ \ifthenelse{\boolean{maledoc}}{apresentado}{apresentada}
2288   \sloppy\nohyphens{{\ ' a} COPPE/UFRJ como parte dos requisitos
2289   necess{\ ' a}rios para a obten{\c c}{\~ a}o do grau de
2290   \local@degname\ em Ci{\^ e}ncias (\@degree)}
2291 }
2292 \newcommand\foreignabstract@toptext{%
2293   \noindent \sloppy\nohyphens{Abstract of \foreign@doctype\ presented to
2294   COPPE/UFRJ as a partial fulfillment of the requirements for the
2295   degree of \foreign@degname\ of Science (\@degree)}
2296 }
2297 %

```

Late-load packages: setspace, hyperref, biblatex

These three packages *must* be the last things the class loads, and in this exact order:

- `setspace` – supplies `\singlespacing`, `\onehalfspacing` and friends. It is loaded here so the line-spacing state it carries around is settled before `hyperref` patches all the cross-reference and TOC plumbing.
- `hyperref` – patches an enormous number of LaTeX internals (`\section`, `\label`, `\ref`, `\bibcite`, `list-of-*` commands, ...). Loading it as late as possible is the standard rule of thumb so it sees the final version of those macros, including the ones the `coppe` class redefines (e.g. `\@chapter`, `\listofquadtros`).
- `biblatex` – must come after `hyperref` so its citation links and back-references hook into `hyperref` correctly; otherwise the `\cite` commands print as plain (non-clickable) text.

```

2298 \RequirePackage{setspace}
2299 % Apply the default line spacing now that setspace is available. The flag was
2300 % set above by \DeclareOption{doublespacing}; absent the option, one-and-a-half.
2301 \if@coppedoublespace\doublespacing\else\onehalfspacing\fi
2302 \RequirePackage{hyperref}
2303 % Bibliography engine: the full coppe BibLaTeX/biber ABNT style, replacing the
2304 % former natbib+bibtex setup. The numbers class option selects the numeric
2305 % citation style; otherwise the author-date default is used. natbib=true keeps
2306 % \citel/\citep available for users migrating from natbib.
2307 \if@coppenumeric
2308   \RequirePackage[backend=biber,datamodel=coppe,%
2309     bibstyle=coppe-numeric,citestyle=coppe-numeric,natbib=true]{biblatex}
2310 \else
2311   \RequirePackage[backend=biber,datamodel=coppe,%
2312     bibstyle=coppe,citestyle=coppe,natbib=true]{biblatex}
2313 \fi
2314 % Reference-list heading "Referencias" (pt) / "References" (en) -- manual2024
2315 % 3.1.4.1 requires the post-textual list to be titled "Referencias", not
2316 % "Bibliografia" (babel's default \bibname). The title is rendered through
2317 % \coppestring so it follows the current main language in BOTH the chapter
2318 % heading and the .toc entry (biblatex's \bibstring proved unreliable across
2319 % the .toc write, and babel's \bibname caption override was equally fragile).
2320 \defbibheading{bibliography}{%
2321   \chapter*{\coppestring{references}}%
2322   \addcontentsline{toc}{chapter}{\coppestring{references}}}
2323 % The catalog page range (bib:begin/bib:end) was written by the old
2324 % thebibliography redefinition, which biblatex's \printbibliography does not
2325 % use. biblatex offers \AtBeginBibliography but no end counterpart, so the
2326 % begin label uses that hook and the end label is emitted by wrapping
2327 % \printbibliography to write it just after the reference list.
2328 \AtBeginBibliography{\coppe@bibBegin}
2329 \let\coppe@orig@printbibliography\printbibliography
2330 \renewcommand\printbibliography[1][1]{%
2331   \coppe@orig@printbibliography[#1]%
2332   \coppe@bibEnd}
2333 \end{class}
2334 \glossary
2335 actual '='
2336 quote '!'

```

```

2337 level '>'
2338 %%%% delim_0    ", p. "
2339 delim_0 "\\dotfill "
2340 lethead_flag 0
2341 headings_flag 0
2342 preamble
2343 "\n\\begin{theglossary}\n  \\makeatletter"
2344 postamble
2345 "\n  \\end{theglossary}\n"
2346 </glossary>

2347 <*class>
2348 \\newcommand{\\annex}{
2349   \\renewcommand{\\appendixname}{\\coppestring{annex}}%
2350   \\appendix
2351   \\@coppeannextrue
2352 }
2353 </class>

```

BibLaTeX style files

Generated from this .dtx by coppe.ins (one docstrip module each).

```

2354 <*dbx>
2355 \\ProvidesFile{coppe.dbx}[2026/05/28 v4.0 CoppeTeX ABNT datamodel]
2356 % Custom entry types beyond biblatex's standard datamodel (games, TV, maps).
2357 % map = cartographic document (NBR 6023 4.2.12); biblatex has no native @map.
2358 % The remaining ABNT "exotic" types (legislation, jurisdiction, legal, patent,
2359 % standard, music, audio, image, artwork, video, dataset, letter, periodical,
2360 % performance) are already in biblatex's standard datamodel.
2361 \\DeclareDatamodelEntrytypes{game,videogame,boardgame,tvshow,map}
2362 % Custom fields.
2363 \\DeclareDatamodelFields[type=field,datatype=literal]{course}      % "(... em <course>)"
2364 \\DeclareDatamodelFields[type=field,datatype=literal]{coppedegree} % degree word
2365 \\DeclareDatamodelFields[type=field,datatype=literal]{platform}   % videogame platform
2366 \\DeclareDatamodelFields[type=field,datatype=literal]{scale}      % map scale (escala)
2367 \\DeclareDatamodelFields[type=list,datatype=literal]{artist}      % game/illustration art
2368 \\DeclareDatamodelFields[type=list,datatype=literal]{director}    % movie/tvshow/video
2369 \\DeclareDatamodelFields[type=list,datatype=literal]{producer}
2370 \\DeclareDatamodelFields[type=field,datatype=literal]{caderno}     % newspaper section
2371 \\DeclareDatamodelFields[type=field,datatype=literal]{relator}     % jurisprudence rapporteur
2372 \\DeclareDatamodelEntryfields{course,coppedegree}
2373 \\DeclareDatamodelEntryfields[game,videogame,boardgame,software,movie,tvshow,video]%
2374   {artist,platform,director,producer,version}
2375 \\DeclareDatamodelEntryfields[map]{scale}
2376 \\DeclareDatamodelEntryfields[article]{caderno}
2377 \\DeclareDatamodelEntryfields[jurisdiction]{relator}
2378 </dbx>
2379 <*bbx>
2380 \\ProvidesFile{coppe.bbx}[2026/05/28 v4.0 CoppeTeX ABNT bibliography style]
2381 % The reference list is single-spaced (ABNT 2.4) via \\singlespacing in
2382 % \\AtBeginBibliography. coppe.cls already loads setspace, but the style must work
2383 % standalone too (e.g. \\usepackage[bibstyle=coppe]{biblatex} in a plain article),
2384 % so load setspace here as well (idempotent).
2385 \\RequirePackage{setspace}

```

```

2386 % ABNT reference list keeps the date at the end (imprenta), same for author-date
2387 % and numeric modes, so we derive from the standard style, not authoryear.
2388 \RequireBibliographyStyle{standard}
2389
2390 % Note: standard style never dashes repeated authors, which already matches the
2391 % post-2020 ABNT rule (repeat the entry in full), so no 'dashed' option is needed.
2392 \ExecuteBibliographyOptions{%
2393   maxbibnames=99,      % ABNT 4.3.1.3: list all authors by default
2394   giveninits=false,
2395   uniquename=false,
2396   uniquelist=false,
2397   labeldateparts=true, % needed for author-date extradate (1999a/1999b)
2398   sorting=nyt,
2399 }
2400
2401 % Language strings (Disponível em / Acesso em / In, etc.)
2402 \DeclareLanguageMapping{brazilian}{brazilian-coppe}
2403 \DeclareLanguageMapping{english}{english-coppe}
2404
2405 % --- Names: all reversed "FAMILY, Given", separated by ";" in the list ---
2406 \DeclareNameAlias{author}{family-given}
2407 \DeclareNameAlias{editor}{family-given}
2408 \DeclareNameAlias{translator}{family-given}
2409 \DeclareNameAlias{default}{family-given}
2410 \DeclareDelimFormat[bib,biblist]{multinamedelim}{\addsemicolon\space}
2411 \DeclareDelimFormat[bib,biblist]{finalnamedelim}{\addsemicolon\space}
2412
2413 % ABNT (NBR 6023 8.1.1.4): a collection entered under the name of its
2414 % organizer prints the kind of responsibility in parentheses after the
2415 % name, abbreviated, lowercase and in the singular even for several
2416 % organizers -- e.g. "SOUSA, A. M. de; OLIVEIRA, L. da C. (org.)". The
2417 % standard style prints ", Org." (comma, no parentheses) when the editor
2418 % takes the author slot, so override editor+others to wrap the role string
2419 % in "(...)". The host position ("In: NAME (org.)") already parenthesises
2420 % the role via coppe:bookeditor below. The role word itself is the
2421 % lowercase, singular "org." (and "ed.", "cur.", ...) set in the language
2422 % packs (brazilian-coppe.lbx et al.).
2423 \renewbibmacro*{editor+others}{%
2424   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
2425     {\printnames{editor}%
2426       \setunit*{\addspace}%
2427       \printtext{\mkbibparens{\usebibmacro{editor+othersstrg}}}%
2428       \clearname{editor}}
2429   {}
2430 % Same parenthesised role for drivers that take the editor through the plain
2431 % editor macro instead of editor+others (e.g. @manual).
2432 \renewbibmacro*{editor}{%
2433   \ifboolexpr{ test \ifuseeditor and not test {\ifnameundef{editor}} }
2434     {\printnames{editor}%
2435       \setunit*{\addspace}%
2436       \printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
2437       \clearname{editor}}
2438   {}
2439

```

```

2440 % Uppercase the surname (family + prefix) only inside the bibliography list;
2441 % citations keep normal caps (post-2020 rule).
2442 % ABNT date (NBR 6023 §4.3.5.5.2): "dia mês. ano" -- without the brazilian "de"
2443 % between the parts (e.g. "4 fev. 1992", "abr. 1996"). #1=year, #2=month, #3=day
2444 % field names; the month stays abbreviated via \mkbibmonth.
2445 \protected\def\coppe@abntdate#1#2#3{%
2446   \iffieldundef{#3}{\stripzeros{\thefield{#3}}\nobreakspace}%
2447   \iffieldundef{#2}{\mkbibmonth{\thefield{#2}}\iffieldundef{#1}{\space}}%
2448   \iffieldundef{#1}{\stripzeros{\thefield{#1}}}%
2449 \AtBeginBibliography{%
2450   \singlespacing % entries single-spaced internally (2.4)
2451   \setlength{\bibitemsep}{\baselineskip}% one blank line between entries (2.4)
2452   \renewcommand*{\mkbibnamefamily}[1]{\MakeUppercase{#1}}%
2453   \renewcommand*{\mkbibnameprefix}[1]{\MakeUppercase{#1}}%
2454   \let\mkbibdatelong\coppe@abntdate % ABNT date format (drop the "de")
2455   \let\mkbibdateshort\coppe@abntdate
2456 }
2457
2458 % --- Titles: bold for whole-document types; plain for parts; entry-by-title
2459 % (no author/editor) -> not bold, first word in CAPS (ABNT 4.2). ---
2460 % \coppeUCtitle uppercases the first SIGNIFICANT word of the title (ABNT 4.2).
2461 % If the title starts with an article (o/a/os/as/um/uma/uns/umas, the/an), the
2462 % article AND the next word are uppercased; otherwise just the first word. The
2463 % delimited macros split off words; \coppeUCkeep strips the trailing-space
2464 % sentinel so no spurious space is left before the following punctuation.
2465 \protected\def\coppeUCtitle#1{\coppeUCtitleA#1 \coppeUCstop}
2466 \protected\def\coppeUCtitleA#1 #2\coppeUCstop{%
2467   \coppeUCifart{#1}
2468   {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCtitleB#2\coppeUCstop}}%
2469   {\MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}%
2470 \protected\def\coppeUCtitleB#1 #2\coppeUCstop{%
2471   \MakeUppercase{#1}\ifblank{#2}{\space\coppeUCkeep#2\coppeUCstop}}
2472 \protected\def\coppeUCkeep#1 \coppeUCstop{#1}
2473 % \coppeUCifart{word}{if-article}{if-not}: case-insensitive membership test.
2474 \protected\def\coppeUCifart#1{%
2475   \begingroup
2476   \lowercase{\def\coppeUC@w{#1}}%
2477   \edef\coppeUC@w{,\coppeUC@w,%}
2478   \expandafter\in@\expandafter{\coppeUC@w}{,a,o,as,os,um,uma,uns,umas,the,an,%}
2479   \ifin@\aftergroup\@firstoftwo\else\aftergroup\@secondoftwo\fi
2480   \endgroup}
2481 \DeclareFieldFormat{title}{%
2482   \ifbool{test}{\ifnameundef{labelname}}{ }
2483   {#1}%
2484   {\mkbibbold{#1}}}
2485 % First word of the title in CAPS for entry-by-title (ABNT 4.2). The titlecase
2486 % format receives the RAW title text (the title format above only sees the
2487 % field-printing commands), so \coppeUCtitle can split off and uppercase the
2488 % first word here.
2489 \DeclareFieldFormat{titlecase}{%
2490   \ifbool{test}{\ifnameundef{labelname}}{ }
2491   {\coppeUCtitle{#1}}%
2492   {#1}}
2493 \DeclareFieldFormat[article,incollection,inbook,inproceedings,suppperiodical,review]{title}{%

```

```

2494 \DeclareFieldFormat[thesis]{title}{\mkbibbold{#1}}
2495 % biblatex's standard style quotes patent (and unpublished) titles; ABNT bolds
2496 % the title of a whole document, so override those back to bold.
2497 \DeclareFieldFormat[patent]{title}{\mkbibbold{#1}}
2498 % Legal documents (NBR 6023 4.2.6): the law's identification/ementa is NOT
2499 % highlighted; the entry leads with the jurisdiction (entity author, CAPS).
2500 \DeclareFieldFormat[legislation,jurisdiction,legal]{title}{#1}
2501 \DeclareFieldFormat[journaltitle]{\mkbibbold{#1}}
2502 \DeclareFieldFormat[booktitle]{\mkbibbold{#1}}
2503 \DeclareFieldFormat[maintitle]{\mkbibbold{#1}}
2504
2505 % --- Pages: ABNT uses "p." (also for ranges); theses use "f." for pagetotal ---
2506 \DeclareFieldFormat[pagetotal]{#1\addnbspace p\adddot}
2507 \DeclareFieldFormat[thesis]{pagetotal}{#1\addnbspace f\adddot}
2508 \DeclareFieldFormat{pages}{p\adddot\addnbspace#1}
2509
2510 % --- Periodical volume / number prefixes ---
2511 \DeclareFieldFormat[article]{volume}{v\adddot\addnbspace#1}
2512 \DeclareFieldFormat[article]{number}{n\adddot\addnbspace#1}
2513
2514 % --- "In:" before host title (incollection/inproceedings/inbook) ---
2515 \renewbibmacro*{in:}{\printtext{\bibstring{in}\addcolon\space}}
2516
2517 % --- Online: "Disponível em: <url>." + "Acesso em: <date>." via lbx strings ---
2518 \DeclareFieldFormat[url]{\bibstring{availablefrom}\addcolon\space\url{#1}}
2519
2520 % --- ABNT article driver: no "In:"; Journal, Local, v., n., p., date, DOI/URL.
2521 % Newspapers (entrysubtype=newspaper, set by the @artigodejornal source map)
2522 % differ: the newspaper name (journaltitle) and city, then either
2523 % "<date>. <caderno>, p. <pages>" (when a caderno/section is given), or
2524 % "p. <pages>, <date>" (page before date, when none),
2525 % per NBR 6023 §4.2.3.6. ---
2526 \DeclareBibliographyDriver{article}{%
2527   \usebibmacro{bibindex}%
2528   \usebibmacro{begentry}%
2529   \usebibmacro{author/translator+others}%
2530   \setunit{\labelnamepunct}\newblock
2531   \usebibmacro{title}%
2532   \newunit\newblock
2533   \printfield{journaltitle}%
2534   \setunit{\addcomma\space}\printlist{location}%
2535   \iffieldequalstr{entrysubtype}{newspaper}
2536     {\iffieldundef{caderno}
2537       {\setunit{\addcomma\space}\printfield{pages}%
2538         \setunit{\addcomma\space}\printdate}%
2539       {\setunit{\addcomma\space}\printdate%
2540         \setunit{\adddot\space}\printfield{caderno}%
2541         \setunit{\addcomma\space}\printfield{pages}}}%
2542   {\setunit{\addcomma\space}\printfield{volume}%
2543     \setunit{\addcomma\space}\printfield{number}%
2544     \setunit{\addcomma\space}\printfield{pages}%
2545     \setunit{\addcomma\space}\printdate}%
2546   \newunit\newblock
2547   \printfield{doi}%

```

```

2548 \newunit\newblock
2549 \usebibmacro{url+urldate}%
2550 \usebibmacro{finentry}}
2551
2552 % --- incollection: ABNT puts the host organizer before the booktitle, as
2553 % "AUTOR. Parte. In: ORGANIZADOR (Org.). Booktitle. Local: Editora, ano,
2554 % p. X-Y." The organizer role string is "Org./Orgs." (see the .lbx). ---
2555 \newbibmacro*{coppe:bookeditor}{%
2556 \ifnameundef{editor}
2557 {}
2558 {\printnames{editor}%
2559 \setunit*{\addspace}\printtext{\mkbibparens{\usebibmacro{editorstrg}}}%
2560 \clearname{editor}%
2561 \setunit{\adddot\space}}}
2562 \newbibmacro*{coppe:partdriver}{%
2563 \usebibmacro{bibindex}\usebibmacro{begentry}%
2564 \usebibmacro{author/translator+others}%
2565 \setunit{\labelnamepunct}\newblock
2566 \usebibmacro{title}%
2567 \newunit\newblock
2568 \usebibmacro{in:}%
2569 \usebibmacro{coppe:bookeditor}%
2570 \printfield{booktitle}%
2571 \newunit\newblock
2572 \printfield{edition}%
2573 \newunit\newblock
2574 \printlist{location}%
2575 \setunit*{\addcolon\space}\printlist{publisher}%
2576 \setunit*{\addcomma\space}\printdate%
2577 \newunit\newblock
2578 \printfield{pages}%
2579 \setunit*{\addcomma\space}\printfield{chapter}%
2580 \newunit\newblock
2581 \usebibmacro{url+urldate}%
2582 \usebibmacro{finentry}}
2583 \DeclareBibliographyDriver{incollection}{\usebibmacro{coppe:partdriver}}
2584 \DeclareBibliographyDriver{inbook}{\usebibmacro{coppe:partdriver}}
2585
2586 % --- type field: resolve a known bibliography string (babel-aware), else literal ---
2587 \DeclareFieldFormat{type}{\ifbibstring{#1}{\bibstring{#1}}{#1}}
2588
2589 % --- Thesis: "<type> (em <course>)" using the custom course field (ABNT). ---
2590 \DeclareFieldFormat{course}{\mkbibparens{\coppestring{in:}\addspace#1}}
2591 \DeclareBibliographyDriver{thesis}{%
2592 \usebibmacro{bibindex}\usebibmacro{begentry}%
2593 \usebibmacro{author}%
2594 \setunit{\labelnamepunct}\newblock
2595 \usebibmacro{title}%
2596 \newunit\newblock
2597 \printfield{type}\setunit*{\addspace}\printfield{course}%
2598 \newunit\newblock
2599 \printlist{institution}%
2600 \setunit*{\addcomma\space}\printlist{location}%
2601 \setunit*{\addcomma\space}\printdate%

```

```

2602 \newunit\newblock
2603 \printfield{pagetotal}%
2604 \newunit\newblock
2605 \printfield{note}%
2606 \newunit\newblock
2607 \usebibmacro{url+urldate}%
2608 \usebibmacro{finentry}}
2609 % --- Games / audiovisual: render the custom fields (artist, platform,
2610 %      director, producer, version) the generic misc driver ignores. ---
2611 \newbibmacro*{coppe:media}{%
2612   \iflistundef{artist}{ }\setunit{\addperiod\space}%
2613   \printtext{\coppestring{art}\addcolon\space}\printlist{artist}}%
2614   \iffieldundef{platform}{ }\setunit{\addperiod\space}%
2615   \printtext{\coppestring{platform}\addcolon\space}\printfield{platform}%
2616   \iflistundef{director}{ }\setunit{\addperiod\space}%
2617   \printtext{\coppestring{directedby}\addcolon\space}\printlist{director}}%
2618   \iflistundef{producer}{ }\setunit{\addperiod\space}%
2619   \printtext{\coppestring{producedby}\addcolon\space}\printlist{producer}}%
2620 \newbibmacro*{coppe:mediadriver}{%
2621   \usebibmacro{bibindex}\usebibmacro{begentry}%
2622   \usebibmacro{author/translator+others}%
2623   \setunit{\labelnamepunct}\newblock
2624   \usebibmacro{title}%
2625   \newunit\newblock
2626   \usebibmacro{coppe:media}%
2627   \newunit\newblock
2628   \printlist{publisher}%
2629   \setunit*{\addcomma\space}\printlist{location}%
2630   \setunit*{\addcomma\space}\printdate%
2631   \newunit\newblock
2632   \printfield{version}%
2633   \newunit\newblock
2634   \printfield{note}%
2635   \newunit\newblock
2636   \usebibmacro{url+urldate}%
2637   \usebibmacro{finentry}}
2638 \DeclareBibliographyDriver{game}{\usebibmacro{coppe:mediadriver}}
2639 \DeclareBibliographyDriver{videogame}{\usebibmacro{coppe:mediadriver}}
2640 \DeclareBibliographyDriver{boardgame}{\usebibmacro{coppe:mediadriver}}
2641 \DeclareBibliographyDriver{movie}{\usebibmacro{coppe:mediadriver}}
2642 \DeclareBibliographyDriver{tvshow}{\usebibmacro{coppe:mediadriver}}
2643 % Cartographic document (map, NBR 6023 4.2.12): like a book (author/title/
2644 % imprenta) plus the mandatory scale ("Escala 1:N").
2645 \DeclareFieldFormat{scale}{\bibstring{coppescale}\adnbspace#1}
2646 \DeclareBibliographyDriver{map}{%
2647   \usebibmacro{bibindex}\usebibmacro{begentry}%
2648   \usebibmacro{author/translator+others}%
2649   \setunit{\labelnamepunct}\newblock
2650   \usebibmacro{title}%
2651   \newunit\newblock
2652   \printlist{location}%
2653   \setunit*{\addcolon\space}\printlist{publisher}%
2654   \setunit*{\addcomma\space}\printdate%
2655   \newunit\newblock

```

```

2656 \printfield{scale}%
2657 \newunit\newblock
2658 \usebibmacro{url+urldate}%
2659 \usebibmacro{finentry}}
2660 % Patent (NBR 6023 4.2.5): inventor. \textbf{title}. Depositante: holder. type
2661 % number. date. The holder (depositante) is printed in normal case, not the
2662 % ALL-CAPS family treatment the bibliography gives the leading author.
2663 \newbibmacro*{coppe:depositor}{%
2664 \ifnameundef{holder}
2665 {}
2666 {\printtext{\bibstring{depositor}\addcolon\space}%
2667 \begingroup
2668 \def\mkbibnamefamily##1{##1}%
2669 \def\mkbibnameprefix##1{##1}%
2670 \printnames{holder}%
2671 \endgroup}}
2672 \DeclareBibliographyDriver{patent}{%
2673 \usebibmacro{bibindex}%
2674 \usebibmacro{begentry}%
2675 \usebibmacro{author/translator+others}%
2676 \setunit{\labelnamepunct}\newblock
2677 \usebibmacro{title}%
2678 \newunit\newblock
2679 \usebibmacro{coppe:depositor}%
2680 \newunit\newblock
2681 \printfield{type}%
2682 \setunit*{\addspace}\printfield{number}%
2683 \newunit\newblock
2684 \printdate%
2685 \newunit\newblock
2686 \usebibmacro{url+urldate}%
2687 \usebibmacro{finentry}}
2688 % Exotic ABNT types that biblatex's standard style leaves without a driver:
2689 % give each a baseline (book-like for published docs: norma/partitura; misc for
2690 % the rest) so they print. Per-type ABNT refinements can follow later.
2691 \DeclareBibliographyDriver{standard}{\usedriver{}{book}}
2692 \DeclareBibliographyDriver{music}{\usedriver{}{book}}
2693 % Legal documents (NBR 6023 4.2.6-4.2.8): jurisdiction/entity (CAPS author),
2694 % the law id + ementa (plain, not bold), then location and date.
2695 \newbibmacro*{coppe:legaldriver}{%
2696 \usebibmacro{bibindex}\usebibmacro{begentry}%
2697 \usebibmacro{author/translator+others}%
2698 \setunit{\labelnamepunct}\newblock
2699 \usebibmacro{title}%
2700 \newunit\newblock
2701 % Jurisprudência: "Relator: <nome>." after the decision title (NBR 6023).
2702 \iffielddundef{relator}{\printtext{Relator\addcolon\space}\printfield{relator}\newunit\ne
2703 \printlist{location}%
2704 \setunit{\addcomma\space}\printdate%
2705 \newunit\newblock
2706 \usebibmacro{url+urldate}%
2707 \usebibmacro{finentry}}
2708 \DeclareBibliographyDriver{legislation}{\usebibmacro{coppe:legaldriver}}
2709 \DeclareBibliographyDriver{jurisdiction}{\usebibmacro{coppe:legaldriver}}

```

```

2710 \DeclareBibliographyDriver{legal}{\usebibmacro{coppe:legaldriver}}
2711 \DeclareBibliographyDriver{audio}{\usedriver{}{misc}}
2712 \DeclareBibliographyDriver{video}{\usedriver{}{misc}}
2713 \DeclareBibliographyDriver{software}{\usedriver{}{misc}}
2714 \DeclareBibliographyDriver{image}{\usedriver{}{misc}}
2715 \DeclareBibliographyDriver{artwork}{\usedriver{}{misc}}
2716 \DeclareBibliographyDriver{performance}{\usedriver{}{misc}}
2717 \DeclareBibliographyDriver{dataset}{\usedriver{}{misc}}
2718
2719 % --- biber source map: Portuguese synonyms (entry types + fields) + thesis family ---
2720 \DeclareSourcemap{%
2721   \maps[datatype=bibtex]{%
2722     % entry-type PT synonyms -> canonical English
2723     \map{step[typesource=livro,typetarget=book]}
2724     \map{step[typesource=artigo,typetarget=article]}
2725     \map{step[typesource=artigodejornal,typetarget=article,final]\step[fieldset=entrysubtyp
2726     \map{step[typesource=partedelivro,typetarget=incollection]}
2727     \map{step[typesource=emlivro,typetarget=inbook]}
2728     \map{step[typesource=verbete,typetarget=inreference]}
2729     \map{step[typesource=anais,typetarget=proceedings]}
2730     \map{step[typesource=trabalhodeevento,typetarget=inproceedings]}
2731     \map{step[typesource=periodico,typetarget=periodical]}
2732     \map{step[typesource=relatorio,typetarget=report]}
2733     \map{step[typesource=relatoriotecnico,typetarget=report]}
2734     \map{step[typesource>manual,typetarget>manual]}
2735     \map{step[typesource=correspondencia,typetarget=letter]}
2736     \map{step[typesource=patente,typetarget=patent]}
2737     \map{step[typesource=diverso,typetarget=misc]}
2738     \map{step[typesource=jogo,typetarget=game]}
2739     \map{step[typesource=jogoeletronico,typetarget=videogame]}
2740     \map{step[typesource=jogodetabuleiro,typetarget=boardgame]}
2741     \map{step[typesource=filme,typetarget=movie]}
2742     \map{step[typesource=serietv,typetarget=tvshow]}
2743     \map{step[typesource=programadetelevisao,typetarget=tvshow]}
2744     % Exotic ABNT types (NBR 6023 4.2.5-4.2.14); targets are biblatex-native
2745     % except map (declared in coppe.dbx).
2746     \map{step[typesource=legislacao,typetarget=legislation]}
2747     \map{step[typesource=atnormativo,typetarget=legislation]}
2748     \map{step[typesource=jurisprudencia,typetarget=jurisdiction]}
2749     \map{step[typesource=documentojuridico,typetarget=legal]}
2750     \map{step[typesource=documentocivil,typetarget=legal]}
2751     \map{step[typesource=norma,typetarget=standard]}
2752     \map{step[typesource=partitura,typetarget=music]}
2753     \map{step[typesource=iconografico,typetarget=image]}
2754     \map{step[typesource=obradearte,typetarget=artwork]}
2755     \map{step[typesource=tridimensional,typetarget=artwork]}
2756     \map{step[typesource=cartografico,typetarget=map]}
2757     \map{step[typesource=mapa,typetarget=map]}
2758     \map{step[typesource=audiovisual,typetarget=video]}
2759     \map{step[typesource=conjuntodedados,typetarget=dataset]}
2760     \map{step[typesource=fasciculo,typetarget=supperperiodical]}
2761     \map{step[typesource=trabalhoapresentado,typetarget=unpublished]}
2762     % thesis family: retype to thesis and preset the (babel-aware) type word
2763     \map{step[typesource=tese,typetarget=thesis]}

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2764 \map{\step[typesource=dissertacao,typetarget=thesis]}
2765 \map{\step[typesource=masterdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
2766 \map{\step[typesource=mscdissertation,typetarget=thesis,final]\step[fieldset=type,fiel
2767 \map{\step[typesource=dissertacaomestrado,typetarget=thesis,final]\step[fieldset=type,fi
2768 \map{\step[typesource=dscthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=d
2769 \map{\step[typesource=tesedoutorado,typetarget=thesis,final]\step[fieldset=type,fieldval
2770 \map{\step[typesource=phdthesis,typetarget=thesis,final]\step[fieldset=type,fieldvalue=p
2771 \map{\step[typesource=tesephd,typetarget=thesis,final]\step[fieldset=type,fieldvalue=phd
2772 \map{\step[typesource=tcc,typetarget=thesis,final]\step[fieldset=type,fieldvalue=tcc]}
2773 % field PT synonyms (unaccented) -> canonical English
2774 \map{\step[fieldsource=periodico,fieldtarget=journaltitle]}
2775 \map{\step[fieldsource=revista,fieldtarget=journaltitle]}
2776 \map{\step[fieldsource=jornal,fieldtarget=journaltitle]}
2777 \map{\step[fieldsource=titulo,fieldtarget=title]}
2778 \map{\step[fieldsource=subtitulo,fieldtarget=subtitle]}
2779 \map{\step[fieldsource=autor,fieldtarget=author]}
2780 \map{\step[fieldsource=editora,fieldtarget=publisher]}
2781 \map{\step[fieldsource=local,fieldtarget=location]}
2782 \map{\step[fieldsource=ano,fieldtarget=year]}
2783 \map{\step[fieldsource=data,fieldtarget=date]}
2784 \map{\step[fieldsource=edicao,fieldtarget=edition]}
2785 \map{\step[fieldsource=paginas,fieldtarget=pages]}
2786 \map{\step[fieldsource=numero,fieldtarget=number]}
2787 \map{\step[fieldsource=tradutor,fieldtarget=translator]}
2788 \map{\step[fieldsource=organizador,fieldtarget=editor]}
2789 \map{\step[fieldsource=instituicao,fieldtarget=institution]}
2790 \map{\step[fieldsource=tipo,fieldtarget=type]}
2791 \map{\step[fieldsource=nota,fieldtarget=note]}
2792 \map{\step[fieldsource=dataacesso,fieldtarget=urldate]}
2793 \map{\step[fieldsource=curso,fieldtarget=course]}
2794 \map{\step[fieldsource=grau,fieldtarget=coppedegree]}
2795 \map{\step[fieldsource=artista,fieldtarget=artist]}
2796 \map{\step[fieldsource=plataforma,fieldtarget=platform]}
2797 \map{\step[fieldsource=diretor,fieldtarget=director]}
2798 \map{\step[fieldsource=produtor,fieldtarget=producer]}
2799 \map{\step[fieldsource=versao,fieldtarget=version]}
2800 \map{\step[fieldsource=escala,fieldtarget=scale]}
2801 }%
2802 }
2803
2804 % --- ABNT list layout: flush left, single spacing, blank line between entries ---
2805 \setlength{\bibhang}{0pt}
2806 \setlength{\bibitemsep}{\baselineskip}
2807
2808 </bbx>
2809 <*cbx>
2810 \ProvidesFile{coppe.cbx}[2026/05/28 v4.0 CoppeTeX ABNT author-date citations]
2811 % Author-date is the CoppeTeX 4.0 default. authoryear-comp gives compressed
2812 % author-year cites with normal-caps names (post-2020 rule) out of the box.
2813 \RequireCitationStyle{authoryear-comp}
2814
2815 % --- NBR 10520:2023 author separator -----
2816 % Parenthetical citations separate authors with ";" -> (Silva; Souza, 2020).
2817 % Narrative citations (\textcite, natbib \citet) keep comma + "e/and" ->

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2818 % Silva, Souza e Oliveira (2020). The reference list keeps its own ";" via
2819 % coppe.bbx ([bib,biblist]); these contextless formats cover the cite side.
2820 \DeclareDelimFormat{multinamedelim}{\addsemicolon\space}
2821 \DeclareDelimFormat{finalnamedelim}{\addsemicolon\space}
2822 \DeclareDelimFormat[textcite]{multinamedelim}{\addcomma\space}
2823 \DeclareDelimFormat[textcite]{finalnamedelim}{\addspace\bibstring{and}\space}
2824
2825 % --- apud: citation of a citation (ABNT NBR 6023 §4.1.2.2 / NBR 10520) -----
2826 % ABNT lists only the work actually CONSULTED. So only the consulted work (#4, a
2827 % .bib key) is cited and enters the reference list; the ORIGINAL work (#2 author,
2828 % #3 year) was not consulted and is given as plain text -- it is never a cite key,
2829 % so it is never added to the bibliography.
2830 % \citetapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2831 % -> Orig Author (year) apud Consulted (yr[, page]) [narrative]
2832 % \citepapud[<page>]{<orig author>}{<orig year>}{<consulted key>}
2833 % -> (Orig Author, year apud CONSULTED, yr[, page]) [parenthetical]
2834 % #1 = optional postnote (page/loc) on the consulted work; "apud" is latin -> italic.
2835 \DeclareRobustCommand{\citetapud}[4][]{%
2836 #2\space{#3}\space\mkbibemph{apud}\space\textcite{#1}{#4}}
2837 \DeclareRobustCommand{\citepapud}[4][]{%
2838 \mkbibparens{#2\addcomma\space#3\space\mkbibemph{apud}\space
2839 \citeauthor{#4}\addcomma\space\citeyear{#4}%
2840 \ifblank{#1}{ }\addcomma\space\mkpageprefix[pagination]{#1}}}}
2841 </cbx>
2842 <*lbxbr>
2843 \ProvidesFile{brazilian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX brazilian strings]
2844 \InheritBibliographyExtras{brazilian}
2845 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
2846 \DeclareBibliographyStrings{%
2847 inherit = {brazilian},
2848 availablefrom = {{Dispon\'}{\i}vel em}{Dispon\'}{\i}vel em}},
2849 urlseen = {{Acesso em}{Acesso em}},
2850 in = {{In}{In}},
2851 editor = {{organizador}{org\adddot}},
2852 editors = {{organizadores}{org\adddot}},
2853 depositor = {{Depositante}{Depositante}},
2854 coppescale = {{Escala}{Escala}},
2855 mscdiss = {{Disserta\c{c}\~{a}o de Mestrado}{Disserta\c{c}\~{a}o de Mestrado}},
2856 dscthesi = {{Tese de Doutorado}{Tese de Doutorado}},
2857 phdthesis = {{Tese de Doutorado}{Tese de Doutorado}},
2858 tcc = {{Trabalho de Conclus\~{a}o de Curso}{Trabalho de Conclus\~{a}o de Curso}}
2859 }
2860 </lbxbr>
2861 <*lbxen>
2862 \ProvidesFile{english-coppe.lbx}[2026/05/28 v4.0 CoppeTeX english strings]
2863 \InheritBibliographyExtras{english}
2864 \NewBibliographyString{availablefrom,mscdiss,dscthesi,tcc,depositor,coppescale}
2865 \DeclareBibliographyStrings{%
2866 inherit = {english},
2867 availablefrom = {{Available from}{Available from}},
2868 urlseen = {{Access on}{Access on}},
2869 in = {{In}{In}},
2870 editor = {{editor}{ed\adddot}},
2871 editors = {{editors}{ed\adddot}},

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2872 depositor      = {{Depositor}}{Depositor}},
2873 coppescale      = {{Scale}}{Scale}},
2874 msdiss          = {{M\addot Sc\addot\ Dissertation}}{M\addot Sc\addot\ Diss\addot}},
2875 dscthesi        = {{D\addot Sc\addot\ Thesis}}{D\addot Sc\addot\ Thesis}},
2876 phdthesis       = {{Ph\addot D\addot\ Thesis}}{Ph\addot D\addot\ Thesis}},
2877 tcc             = {{Undergraduate Thesis}}{Undergraduate Thesis}},
2878 }
2879 </lboxen>
2880 <*lboxes>
2881 \ProvidesFile{spanish-coppe.lbx}[2026/05/28 v4.0 CoppeTeX spanish strings]
2882 \InheritBibliographyExtras{spanish}
2883 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale}
2884 \DeclareBibliographyStrings{%
2885   inherit        = {spanish},
2886   availablefrom = {{Disponible en}}{Disponible en}},
2887   urlseen       = {{Consultado el}}{Consultado el}},
2888   in            = {{En}}{En}},
2889   editor        = {{editor}}{ed\addot}},
2890   editors       = {{editores}}{ed\addot}},
2891   depositor     = {{Solicitante}}{Solicitante}},
2892   coppescale    = {{Escala}}{Escala}},
2893   msdiss        = {{Tesis de Maestr'\{i\}a}}{Tesis de Maestr'\{i\}a}},
2894   dscthesi      = {{Tesis Doctoral}}{Tesis Doctoral}},
2895   phdthesis     = {{Tesis Doctoral}}{Tesis Doctoral}},
2896   tcc           = {{Trabajo de Fin de Carrera}}{Trabajo de Fin de Carrera}},
2897 }
2898 </lboxes>
2899 <*lboxfr>
2900 \ProvidesFile{french-coppe.lbx}[2026/05/28 v4.0 CoppeTeX french strings]
2901 \InheritBibliographyExtras{french}
2902 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale}
2903 \DeclareBibliographyStrings{%
2904   inherit        = {french},
2905   availablefrom = {{Disponible \{a\}}{Disponible \{a\}}},
2906   urlseen       = {{Consult\{e\} le}}{Consult\{e\} le}},
2907   in            = {{Dans}}{Dans}},
2908   editor        = {{\{\'editeur\}}{\{\'ed\addot\}}},
2909   editors       = {{\{\'editeurs\}}{\{\'ed\addot\}}},
2910   depositor     = {{D\{\'eposant\}}{D\{\'eposant\}}},
2911   coppescale    = {{\{\'Echelle\}}{\{\'Echelle\}}},
2912   msdiss        = {{M\{\'emoire de Master\}}{M\{\'emoire de Master\}}},
2913   dscthesi      = {{Th\{\'ese de Doctorat\}}{Th\{\'ese de Doctorat\}}},
2914   phdthesis     = {{Th\{\'ese de Doctorat\}}{Th\{\'ese de Doctorat\}}},
2915   tcc           = {{M\{\'emoire de fin d\{\'etudes\}}{M\{\'emoire de fin d\{\'etudes\}}},
2916 }
2917 </lboxfr>
2918 <*lboxit>
2919 \ProvidesFile{italian-coppe.lbx}[2026/05/28 v4.0 CoppeTeX italian strings]
2920 \InheritBibliographyExtras{italian}
2921 \NewBibliographyString{availablefrom,msdiss,dscthesi,tcc,depositor,coppescale}
2922 \DeclareBibliographyStrings{%
2923   inherit        = {italian},
2924   availablefrom = {{Disponibile da}}{Disponibile da}},
2925   urlseen       = {{Consultato il}}{Consultato il}},

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2926 in          = {{In}{In}},
2927 editor       = {{curatore}{cur\adddot}},
2928 editors      = {{curatori}{cur\adddot}},
2929 depositor    = {{Depositante}{Depositante}},
2930 coppescale   = {{Scala}{Scala}},
2931 msdiss       = {{Tesi di Laurea Magistrale}{Tesi di Laurea Magistrale}},
2932 dscthesi     = {{Tesi di Dottorato}{Tesi di Dottorato}},
2933 phdthesis    = {{Tesi di Dottorato}{Tesi di Dottorato}},
2934 tcc         = {{Tesi di Laurea Triennale}{Tesi di Laurea Triennale}},
2935 }
2936 </lboxit>
2937 <*\langes>
2938 %% coppe-lang-spanish.def -- CoppeTeX 4.x Spanish class-string pack.
2939 %% Loaded by coppe.cls when \documentclass[spanish]{coppe} (or any other
2940 %% spanish slot) is active. Populates the dispatcher table for the
2941 %% spanish language and registers the biblatex language mapping.
2942 %% Institutional names (universityname/cityname/statename/countryname)
2943 %% are deliberately kept Portuguese: the COPPE/UFRJ cover and folha de
2944 %% rosto are a Brazilian institutional template that does not translate.
2945 \ProvidesFile{coppe-lang-spanish.def}[2026/05/28 v4.0 CoppeTeX spanish class strings]
2946 \copperdefstring{spanish}{appendix}      {Ap\'endice}
2947 \copperdefstring{spanish}{annex}         {Anexo}
2948 \copperdefstring{spanish}{frame}         {Cuadro}
2949 \copperdefstring{spanish}{listframe}     {Lista de Cuadros}
2950 \copperdefstring{spanish}{source}        {Fuente}
2951 \copperdefstring{spanish}{program}       {Programa}
2952 \copperdefstring{spanish}{listprogram}   {Lista de Programas}
2953 \copperdefstring{spanish}{references}    {Referencias}
2954 \copperdefstring{spanish}{in:}           {en}
2955 \copperdefstring{spanish}{art}           {Arte}
2956 \copperdefstring{spanish}{platform}      {Plataforma}
2957 \copperdefstring{spanish}{directedby}    {Direcci\'on}
2958 \copperdefstring{spanish}{producedby}    {Producci\'on}
2959 \copperdefstring{spanish}{listabbreviation}{Lista de Abreviaturas y Siglas}
2960 \copperdefstring{spanish}{listsymbol}    {Lista de S\'imbolos}
2961 \copperdefstring{spanish}{glossary}      {Glosario}
2962 \copperdefstring{spanish}{advisor}       {Director}
2963 \copperdefstring{spanish}{advisors}      {Directores}
2964 \copperdefstring{spanish}{dept}          {Programa}
2965 \copperdefstring{spanish}{approvedmale}  {Examinado por}
2966 \copperdefstring{spanish}{approvedfemale}{Examinada por}
2967 \copperdefstring{spanish}{universityname}{Universidade Federal do Rio de Janeiro}
2968 \copperdefstring{spanish}{cityname}     {Rio de Janeiro}
2969 \copperdefstring{spanish}{statename}    {RJ}
2970 \copperdefstring{spanish}{countryname}  {Brasil}
2971 \copperdefstring{spanish}{monthname1}   {enero}
2972 \copperdefstring{spanish}{monthname2}   {febrero}
2973 \copperdefstring{spanish}{monthname3}   {marzo}
2974 \copperdefstring{spanish}{monthname4}   {abril}
2975 \copperdefstring{spanish}{monthname5}   {mayo}
2976 \copperdefstring{spanish}{monthname6}   {junio}
2977 \copperdefstring{spanish}{monthname7}   {julio}
2978 \copperdefstring{spanish}{monthname8}   {agosto}
2979 \copperdefstring{spanish}{monthname9}   {septiembre}

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2980 \copperdefstring{spanish}{monthname10}{octubre}
2981 \copperdefstring{spanish}{monthname11}{noviembre}
2982 \copperdefstring{spanish}{monthname12}{diciembre}
2983 \copperdefstring{spanish}{algorithm2eopt}{spanish}
2984 \copperdefstring{spanish}{babelname} {spanish}
2985 %% biblatex is loaded by coppe.cls AFTER this .def file is read; defer the
2986 %% language-mapping registration until \begin{document} so biblatex's
2987 %% \DeclareLanguageMapping is available.
2988 \AtBeginDocument{\DeclareLanguageMapping{spanish}{spanish-coppe}}
2989 
2990 (*langfr)
2991 %% coppe-lang-french.def -- CoppeTeX 4.x French class-string pack.
2992 %% See coppe-lang-spanish.def for the structure. Institutional names
2993 %% remain Portuguese; only document-content strings translate.
2994 \ProvidesFile{coppe-lang-french.def}[2026/05/28 v4.0 CoppeTeX french class strings]
2995 \copperdefstring{french}{appendix} {Annexe}
2996 \copperdefstring{french}{annex} {Annexe}
2997 \copperdefstring{french}{frame} {Encadr\'e}
2998 \copperdefstring{french}{listframe} {Liste des encadr\'es}
2999 \copperdefstring{french}{source} {Source}
3000 \copperdefstring{french}{program} {Programme}
3001 \copperdefstring{french}{listprogram} {Liste des programmes}
3002 \copperdefstring{french}{references} {R\'ef\'erences}
3003 \copperdefstring{french}{in:} {dans}
3004 \copperdefstring{french}{art} {Illustration}
3005 \copperdefstring{french}{platform} {Plateforme}
3006 \copperdefstring{french}{directedby} {R\'ealis\'e par}
3007 \copperdefstring{french}{producedby} {Produit par}
3008 \copperdefstring{french}{listabbreviation} {Liste des abr\'eviations et acronymes}
3009 \copperdefstring{french}{listsymbol} {Liste des symboles}
3010 \copperdefstring{french}{glossary} {Glossaire}
3011 \copperdefstring{french}{advisor} {Directeur}
3012 \copperdefstring{french}{advisors} {Directeurs}
3013 \copperdefstring{french}{dept} {Programme}
3014 \copperdefstring{french}{approvedmale} {Examin\'e par}
3015 \copperdefstring{french}{approvedfemale} {Examin\'ee par}
3016 \copperdefstring{french}{universityname} {Universidade Federal do Rio de Janeiro}
3017 \copperdefstring{french}{cityname} {Rio de Janeiro}
3018 \copperdefstring{french}{statename} {RJ}
3019 \copperdefstring{french}{countryname} {Brasil}
3020 \copperdefstring{french}{monthname1} {janvier}
3021 \copperdefstring{french}{monthname2} {f\'evrier}
3022 \copperdefstring{french}{monthname3} {mars}
3023 \copperdefstring{french}{monthname4} {avril}
3024 \copperdefstring{french}{monthname5} {mai}
3025 \copperdefstring{french}{monthname6} {juin}
3026 \copperdefstring{french}{monthname7} {juillet}
3027 \copperdefstring{french}{monthname8} {ao\~ut}
3028 \copperdefstring{french}{monthname9} {septembre}
3029 \copperdefstring{french}{monthname10}{octobre}
3030 \copperdefstring{french}{monthname11}{novembre}
3031 \copperdefstring{french}{monthname12}{d\'ecembre}
3032 \copperdefstring{french}{algorithm2eopt}{french}
3033 \copperdefstring{french}{babelname} {french}

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3034 \AtBeginDocument{\DeclareLanguageMapping{french}{french-coppe}}
3035 \</langfr>
3036 \<*langit>
3037 %% coppe-lang-italian.def -- CoppeTeX 4.x Italian class-string pack.
3038 %% See coppe-lang-spanish.def for the structure. The algorithm2e
3039 %% language option for Italian is spelled "italiano" -- that's the
3040 %% spelling algorithm2e ships, even though babel calls the language
3041 %% "italian".
3042 \ProvidesFile{coppe-lang-italian.def}[2026/05/28 v4.0 CoppeTeX italian class strings]
3043 \copperdefstring{italian}{appendix}      {Appendice}
3044 \copperdefstring{italian}{annex}         {Allegato}
3045 \copperdefstring{italian}{frame}         {Riquadro}
3046 \copperdefstring{italian}{listframe}     {Elenco dei riquadri}
3047 \copperdefstring{italian}{source}        {Fonte}
3048 \copperdefstring{italian}{program}       {Programma}
3049 \copperdefstring{italian}{listprogram}    {Elenco dei programmi}
3050 \copperdefstring{italian}{references}     {Riferimenti}
3051 \copperdefstring{italian}{in:}           {in}
3052 \copperdefstring{italian}{art}           {Illustrazione}
3053 \copperdefstring{italian}{platform}      {Piattaforma}
3054 \copperdefstring{italian}{directedby}    {Regia di}
3055 \copperdefstring{italian}{producedby}    {Prodotto da}
3056 \copperdefstring{italian}{listabbreviation}{Elenco di abbreviazioni e sigle}
3057 \copperdefstring{italian}{listsymbol}    {Elenco dei simboli}
3058 \copperdefstring{italian}{glossary}      {Glossario}
3059 \copperdefstring{italian}{advisor}       {Relatore}
3060 \copperdefstring{italian}{advisors}      {Relatori}
3061 \copperdefstring{italian}{dept}          {Programma}
3062 \copperdefstring{italian}{approvedmale}  {Esaminato da}
3063 \copperdefstring{italian}{approvedfemale}{Esaminata da}
3064 \copperdefstring{italian}{universityname}{Universidade Federal do Rio de Janeiro}
3065 \copperdefstring{italian}{cityname}      {Rio de Janeiro}
3066 \copperdefstring{italian}{statename}     {RJ}
3067 \copperdefstring{italian}{countryname}   {Brasil}
3068 \copperdefstring{italian}{monthname1}    {gennaio}
3069 \copperdefstring{italian}{monthname2}    {febbraio}
3070 \copperdefstring{italian}{monthname3}    {marzo}
3071 \copperdefstring{italian}{monthname4}    {aprile}
3072 \copperdefstring{italian}{monthname5}    {maggio}
3073 \copperdefstring{italian}{monthname6}    {giugno}
3074 \copperdefstring{italian}{monthname7}    {luglio}
3075 \copperdefstring{italian}{monthname8}    {agosto}
3076 \copperdefstring{italian}{monthname9}    {settembre}
3077 \copperdefstring{italian}{monthname10}   {ottobre}
3078 \copperdefstring{italian}{monthname11}   {novembre}
3079 \copperdefstring{italian}{monthname12}   {dicembre}
3080 \copperdefstring{italian}{algorithm2eopt}{italiano}
3081 \copperdefstring{italian}{babelname}     {italian}
3082 \AtBeginDocument{\DeclareLanguageMapping{italian}{italian-coppe}}
3083 \</langit>
3084 \<*numbbx>
3085 \ProvidesFile{coppe-numeric.bbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric bibliography]
3086 % Numeric reference list for the ABNT numeric system (NBR 10520). It reuses the
3087 % full coppe ABNT style (drivers, CAPS surnames, bold titles, "In:", p./f.,

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3088 % the PT->EN source map and the language mappings) and only adds the [n] list
3089 % label and the numeric sorting on top.
3090 \RequireBibliographyStyle{coppe}
3091
3092 % ABNT numeric system lists references in the order they are first cited
3093 % (NBR 10520), i.e. unsorted. Override with sorting=nty for an alphabetical
3094 % numbered list. 'labelnumber' generates the [n] field.
3095 \ExecuteBibliographyOptions{labelnumber,sorting=none}
3096
3097 % Bracketed labels, both in the list and for any shorthand entries.
3098 \DeclareFieldFormat{labelnumberwidth}{\mkbibbrackets{#1}}
3099 \DeclareFieldFormat{shorthandwidth}{\mkbibbrackets{#1}}
3100
3101 % Reference list driven by the [n] label (copied from biblatex's numeric.bbx),
3102 % keeping the ABNT blank line between entries via \bibitemsep set in coppe.bbx.
3103 \defbibenvironment{bibliography}
3104 {
  \list
3105   {\printtext[labelnumberwidth]{%
3106     \printfield{labelprefix}%
3107     \printfield{labelnumber}}}
3108   {\setlength{\labelwidth}{\labelnumberwidth}%
3109     \setlength{\leftmargin}{\labelwidth}%
3110     \setlength{\labelsep}{\biblabelsep}%
3111     \addtolength{\leftmargin}{\labelsep}%
3112     \setlength{\itemsep}{\bibitemsep}%
3113     \setlength{\parsep}{\bibparsep}}%
3114   \renewcommand*{\makelabel}[1]{\hss#1}}
3115 {\endlist}
3116 {\item}
3117
3118 </numbbx>
3119 <*numcbx>
3120 \ProvidesFile{coppe-numeric.cbx}[2026/05/28 v4.0 CoppeTeX ABNT numeric citations]
3121 % Numeric citation system (ABNT NBR 10520): in-text references are bracketed
3122 % numbers [1], compressed and sorted within a single cite group. This is the
3123 % opt-in alternative to the author-date default (coppe.cbx); it pairs with
3124 % bibstyle=coppe-numeric. With the biblatex natbib option, \citep -> [1] and
3125 % \citet -> Author [1].
3126 \RequireCitationStyle{numeric-comp}
3127 </numcbx>

```

10.2 The bibliographic databases

The two BibLaTeX/biber databases shipped with the class — `example.bib` (the sample database cited by the example monograph) and `coppe.bib` (the references of this manual) — are generated from this source too, so the whole distribution still has a single editable origin. They live inside an `\iffalse` block below: the documentation driver never typesets them, and only `coppe.ins` extracts them, through the `examplebib` and `coppebib` docstrip targets. Edit the entries here, never the generated `.bib` files.

Acknowledgments

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Numbers written in *italic* refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in roman refer to the code lines where the entry is used.

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v0.1

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v0.2	General: Unification of the code for the list of symbols and abbreviations.	1	v3.4	General: Some examples for figures, tables, longtables, etc. . .
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v0.5	abstract: Changed from macro to environment.	51	v4.0	General: Catalog-label writers (coppe@mainBegin, coppe@bibBegin, coppe@bibEnd, coppe@hasLof) made language-agnostic: detokenize wrap on label names protects ':' and '.' from babel shorthand activation, and the raw etex primitives romannumeral/number replace LaTeX's roman/arabic to bypass spanish-babel's small-caps @roman redefinition that would otherwise crash pdftex with "Unbalanced write command" on every es-main document.
	\backmatter: Added mainmatter pages counter.	45		1
	foreignabstract: Changed from macro to environment.	52		Class-wide string dispatcher (copperdefstring, copperstring, coppemainstring, coppeforeignstring) replaces scattered iflanguage and if english branches; pt/en outputs remain byte-identical. . .
v1.0	General: First COPPETEX release. . .	1		1
v2.0	General: COPPETEX release 2.0. . .	1		Multilingual model: three fixed slots (main, foreign, optional third); five built-in language options (brazilian, english, spanish, french, italian); plug-in mechanism for any other Babel language via coppe-lang-lang.def plus lang-coppe.lbx.
				1
v2.1	General: COPPETEX release 2.1: Matching the new rules.	1		New user API: titlein, usecoppelanguage, brazilianabstract environment. . .
	\advisor: Advisors, co-advisors, co-co-advisors, etc., all of them are simply considered advisors. . .	43		1
v2.2	General: Matching new guidelines, including new logo.	1		\coppetexfinalpage: Initial coppetexfinalpage colophon page.
				46
v2.2.2	General: Fixed some text constants in .bib and documented it here. . .	1		
v3.0	General: Added support for monographs in English.	41		
	New approval page layout. . . .	48		
	\department: Added new course on Nanotechnology.	42		
	\examiner: Examiners expansion without degree.	43		
v3.1	\department: Included a sort key in symb	42		
v3.2	General: Fixed version problem between cls and dtx.	1		
v3.3	General: Extend abbrev to work like symb and accept a sorting key . .	1		